

What Adapter Ensembling Actually Buys: A Pre-Registered Audit Against a Stronger Single Adapter

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Abstract

Parameter-efficient fine-tuning leaves practitioners with populations of adapters, and a large literature reports that combining them helps. We ask what combining them buys once the single-adapter baseline is strong. We pre-register and release ADAPTERFRONTIER, a 1,028-cell corpus (67 pools, 13 base models, 8 tasks) registered at git SHA 02a5a6bc *before any adapter was trained*¹, and adjudicate every (pool, method, baseline, metric) cell with paired bootstrap CIs and per-arm BH-FDR control. The single-adapter baseline is deliberately strong: a median $1.45\times$ the pool’s training compute plus a hyperparameter search the ensemble arm never gets. Against it, encoder ensembling **never significantly improves accuracy** (0 of 92 clean cells; 31.5% significantly *worse*), while decoder ensembling **sometimes does** (14 of 100 cells, none worse, all on MNLI). Both families improve ECE (51/92, 57/100), but the single adapter was never calibrated: on the encoder pools where we can run the control, **one held-out temperature erases the advantage** (ensemble better-calibrated in 4/6 pools $\rightarrow$ 2/6). On encoders, calibrate one adapter rather than serve N ; on decoders, the pool buys a small, task-concentrated gain at $N\times$ compute. We then price three evaluation shortcuts: selecting on test is worth +1.12pp to whichever arm may peek; on the same 48 pools, comparing against the pool’s best member instead of a compute-matched single triples SUPPORTED (7.3% $\rightarrow$ 22.4%); and a -0.012pp “no effect” task mean conceals 31 significant cells. We release the cells, per-example predictions and adjudication code so new methods can be compared without retraining.

¹Cell count: $1,028 = 827 \text{ v1.0 main-pool cells } (55 \text{ pools } \times \text{ensemble methods } \times \{\text{best_of_n, n_rank}\} \times \{\text{accuracy, ECE}\}) + 100 \text{ v1.1 HellaSwag multichoice cells } + 100 \text{ baseline self-comparison cells } + 1 \text{ Phase-0 smoke-test cell}$. The 2 v1.1 GSM8K and 6 v1.2 supplementary pools (App. K) are reported separately in Table 9 and are not in the headline 1,028.

1 Introduction

Parameter-efficient fine-tuning (PEFT) with LoRA-style adapters (Hu et al., 2022) has made it routine to keep dozens of task-specific adapters around: serving stacks such as vLLM and S-LoRA (Kwon et al., 2023; Sheng et al., 2024) route per request to one of N adapters at negligible extra cost, and frameworks such as LoRAHub (Huang et al., 2024) target multi-adapter composition directly. The practitioner’s question is whether *combining* those adapters — averaging weights, voting on labels, or routing through them — beats spending the same budget on a single adapter. A large literature reports that it does. We ask a prior question: *how much of that reported benefit survives a strict evaluation protocol?*

Answering this requires comparisons that were fixed before the data was seen, because the choices involved — which single adapter counts as the baseline, how it was selected, at what budget, and on which benchmark — are exactly the choices that determine the answer. We therefore pre-registered ADAPTERFRONTIER at git SHA 02a5a6bc *before any adapter was trained*, and adjudicate each (pool, method, baseline, metric) cell independently with paired bootstrap CIs, permutation or bootstrap p -values, and Benjamini–Hochberg FDR per metric (Benjamini and Hochberg, 1995).

A strong baseline, and a family split. The comparison that decides this question is the single adapter you would train instead. We make that competitor deliberately strong: our pre-registered `n_rank` baseline spends the pool’s budget on adapter rank, and measured from the released manifests it in fact received a median $1.45\times$ the pool’s training wall-clock (max $2.96\times$; 47 of 48 pool-baseline pairs favour the baseline) together with a $\text{rank}\times\text{epoch}\times\text{seed}$ search selected on held-out data that the ensemble arm never receives. It also serves one adapter per example against the ensem-

ble’s N . It is therefore stronger than a matched competitor, and that cuts against us: a null against a better-resourced single is biased towards the null. The null does not weaken where the advantage is smallest (§4), but we state it against a single given *at least* the pool’s budget. Against it, on the clean encoder slice **no cell shows a significant accuracy gain** (0 of 92, identically pre- and post-FDR) while 31.5% are significantly reversed (mean $\Delta = -1.27\text{pp}$). On the 25 decoder pools, against baselines with the same advantage, the answer differs: 14 of 100 cells SUPPORTED (10/96 without one baseline that failed to train), none reversed, mean $+0.56\text{pp}$ — narrow, all on MNLI, but real. Our pre-registration declared accuracy and calibration co-equal arms, and on *the same* 92 cells ensembling improves ECE in 51. We read that as a dissociation until we noticed the comparison was not fair the way the accuracy arm’s was: that single adapter is chosen by validation *accuracy* and never calibrated. One temperature fitted on held-out data takes the ensemble from better-calibrated in 4/6 encoder pools to 2/6 (§4). The honest statement is narrower and more useful: against a single given the pool’s budget, an encoder population buys neither accuracy nor reliability that a scalar cannot, and a decoder population buys a small accuracy gain at $N \times$ compute.

Three shortcuts move the verdict. That encoder null coexists with a literature reporting gains, so we measure what closes the gap (§5). Selecting the “best single” on the test split is worth $+1.12\text{pp}$ on average to whichever arm is allowed to peek, and inflates 82% of pools. On the same 48 pools, comparing against the pool’s best member rather than a compute-matched single triples the SUPPORTED rate (7.3% $\rightarrow$ 22.4%) and cuts REVERSED from 15.1% to 3.6%. Reporting one mean per task hides the structure entirely: MNLI’s -0.012pp average reads as “no effect” while concealing 31 significant cells pointing in both directions. Each shortcut is individually defensible-looking; together they are enough to flip the sign of the conclusion.

Contributions. (i) **A pre-registered corpus and adjudication framework:** 67 adapter pools and 1,028 cell JSONs with per-cell CIs, p -values and FDR-corrected verdicts, plus the locked pre-registration and amendment ledger; new methods are compared by submitting per-example predictions, with no retraining. (ii) **A family split under a strong baseline, and the control that explains**

the calibration gain: against a single adapter that received more training compute and a hyperparameter search, PEFT ensembling never significantly improves encoder accuracy (0/92) and frequently harms it, while decoder ensembling wins 14/100 cells and loses none. Both improve ECE, but that comparison omits post-hoc calibration; on encoders, supplying it removes most of the effect, which we trace to confidence shrinkage and replace with a single scalar. (iii) **A protocol-sensitivity analysis** pricing three shortcuts that each move the verdict (Table 2, Fig. 1). (iv) **Negative results reported as such:** the diversity–quality frontier predicts accuracy deltas out of pool (corpus-wide LOPO-CV $R^2 = 0.597$ over 48 pools, 95% CI $[+0.30, +0.71]$; encoder-only $R^2 = 0.616$ over 23 pools; provisional, since its prediction-space features were computed partly on the test split, L1c) but on encoders, where those deltas are predominantly negative, it predicts how much a pool stands to *lose*, and it is not a proxy for the encoder/decoder split (family alone: $R^2 = +0.20$); it does not extend to calibration, and three candidate mechanisms are tested and refuted (App. Y).

Descriptive regularities. Five rules D1–D5 (App. Q) summarise where cells differ; derived and evaluated on the same corpus, they are an *oracle upper bound*, not a policy (L11).

2 Related work

PEFT and adapter ensembling. The adapter line of work begins with bottleneck adapters (Houlsby et al., 2019) and the AdapterHub framework (Pfeiffer et al., 2020); LoRA (Hu et al., 2022), DoRA (Liu et al., 2024), PiSSA (Meng et al., 2024), AdaLoRA (Zhang et al., 2023), and VeRA (Kopiczko et al., 2024) make many low-rank adapters per task practical. Combination strategies split into three families: *prediction-space* (soft/hard voting, logit averaging (Lakshminarayanan et al., 2017), including ensembles of LoRA adapters on LLMs (Wang et al., 2023a)); *weight-space* (model soups (Wortsman et al., 2022), TIES (Yadav et al., 2023), DARE (Yu et al., 2024)); and *routing / mixture-of-adapters* (AdapterFusion (Pfeiffer et al., 2021) composes via attention; LoRAHub (Huang et al., 2024) optimizes a continuous weight vector; AdaMix (Wang et al., 2022) mixes multiple LoRA experts inside training; MoLE (Wu et al., 2024) and MoLoRA (Caccia et al., 2023) learn routing over adapter experts;

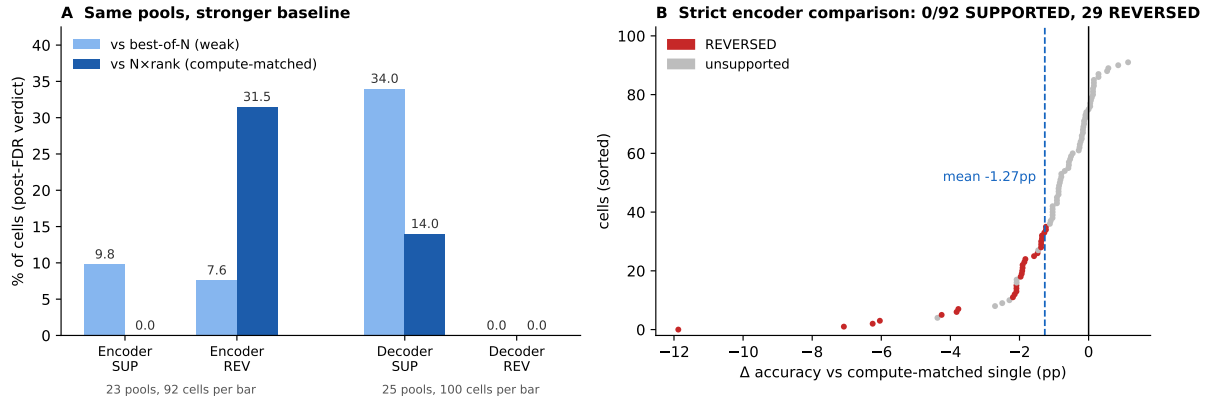


Figure 1: **The apparent ensembling benefit depends on the baseline.** (A) The same 48 pools, adjudicated against the pool’s own best member (`best_of_n`, light) and against the pre-registered compute-matched single (`n_rank`, dark); every pair of bars shares its cells. On encoders SUPPORTED falls from 9.8% to 0% and REVERSED rises from 7.6% to 31.5%; on decoders SUPPORTED falls from 34% to 14% and nothing is REVERSED under either baseline. (B) All 92 strict encoder cells, sorted, colored by post-FDR verdict: the mass lies left of zero (mean -1.27pp), 29 are REVERSED and none SUPPORTED.

HydraLoRA (Tian et al., 2024) uses asymmetric multi-head LoRA; MultiLoRA (Wang et al., 2023b) reduces interference among co-trained LoRAs). Our adjudication framework treats each of these methods as a *plug-in* that produces per-example test predictions and can be added to our cell corpus without retraining the pools. We implement and report prediction-space methods directly (`soft_vote`, `logit_avg`, `majority_vote`, `greedy_soup` — which greedily adds members on `val_selection` and averages their logits, the prediction-space analogue of a greedy soup — plus a prediction-space `uniform_soup proxy` that averages per-adapter logits); weight-space TIES/DARE/uniform-merge implementations live in `weight_space_merge.py` and are reported as a separate family in App. C. The cell JSONs include the input schema needed to evaluate routing/MoE methods (per-pool logits cache).

Pre-registration in ML. Pre-registration is standard in clinical trials and increasingly common in cognitive science but rare in ML (Munafò et al., 2017; Loken and Gelman, 2017); Hofman et al. (2021) argue for tighter coupling of explanation and prediction in computational social science, an argument that translates directly to empirical NLP — Bowman and Dahl (2021) make a closely related case that NLU benchmark results need pre-committed protocols if they are to drive scientific (rather than leaderboard) progress. Dehghani et al. (2022) further argue that compute-matched comparisons are essential to honest efficiency claims — a

principle we apply across every cell. We adopt pre-registration here: the protocol, splits, baselines, and statistical tests were locked at git SHA 02a5a6bc before any pool was trained.

Calibration of fine-tuned LMs. ECE in its modern form was introduced by Naeini et al. (2015); the equal-mass 15-bin histogram became standard via Guo et al. (2017), with refinements such as soft calibration objectives (Karandikar et al., 2021). Nixon et al. (2019) survey ECE estimator bias under varying bin counts and small- N regimes; Minderer et al. (2021) re-examine calibration of modern (post-2020) architectures and find that distribution-shift calibration does not always follow in-distribution calibration. Deep ensembles (Lakshminarayanan et al., 2017) reliably lower ECE on classifiers, but Ashukha et al. (2020) argue that in-domain uncertainty comparisons, ensembles included, should be made after temperature scaling, since an uncalibrated baseline confounds them, and Ovadia et al. (2019) find that temperature scaling’s in-distribution gains do not survive dataset shift, where ensembles hold up best. Our recalibration control applies the first point to PEFT pools; the second bounds it (L13). We extend this measurement to PEFT pools and report Brier, NLL, MCE alongside ECE, with paired bootstrap CIs; per-token logprob ECE for generative tasks (App I) is a non-standard adaptation we explicitly document.

Efficient multi-adapter serving. S-LoRA (Sheng et al., 2024) and Punica (Chen et al., 2024) (the latter integrated in vLLM (Kwon et al.,

2023)) enable thousands of LoRA adapters to share a single base-model forward; we measure the runtime cost of cell-by-cell ensembling under this serving regime.

When reported gains do not survive fair comparison. Several fields have found that a literature of reported improvements shrinks or disappears once baselines are tuned and budgets are matched: GAN variants become indistinguishable under equal hyperparameter search (Lucic et al., 2018), deep metric-learning gains are “marginal at best” under controlled evaluation (Musgrave et al., 2020), and most neural recommenders are beaten by simple heuristics (Ferrari Dacrema et al., 2019). The recurring cause is not fraud but protocol: an under-tuned or under-resourced baseline, a selection step that sees the test set, or an aggregate score that hides variance (Dodge et al., 2019; Reimers and Gurevych, 2017). NLP has documented the same failure modes for benchmarking practice (Bowman and Dahl, 2021) and for efficiency claims specifically, where comparisons that do not fix the compute budget can invert conclusions (Dehghani et al., 2022). Two literatures bear directly on our question. PEFT evaluation has already had its reality check: Chen et al. (2022) find validation and testing practice across PEFT studies unreliable once method instability is accounted for. And *ensemble versus one bigger model at matched cost* is an open dispute rather than a naive assumption: Abe et al. (2022) find a single larger network replicates ensemble gains with diversity contributing little, Kondratyuk et al. (2020) conclude the opposite at matched FLOPs, Lobacheva et al. (2020) show the optimum depends on budget, and Theisen et al. (2023) predict the gain from the disagreement-to-error ratio. Our encoder result is the PEFT instance of the Abe et al. side and our decoder result leans the other way; our frontier is consistent with Theisen et al.. We claim priority for neither. Ours is the pre-registration before training, the cell-level adjudication, and the per-shortcut price in percentage points (§5).

Benchmarks for NLP reasoning. HellaSwag (Zellers et al., 2019), GSM8K (Cobbe et al., 2021), and the GLUE-family tasks (Wang et al., 2018) are scored by our own evaluators, not by lm-evaluation-harness (Gao et al., 2023), though we follow its length-normalised log-likelihood convention on HellaSwag. We use them as cells, not as headline numbers.

3 Benchmark composition

3.1 Pool types

Per (model, task) cell:

- **Pool-A (seed-only):** 20 LoRA adapters with fixed ($r = 8, \alpha = 16, lr = 3e-4, target = qv$), seed varied across $\{11, 21, \dots, 201\}$.
- **Pool-B (HP-diverse):** 24 adapters from $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{1e-4, 3e-4\}$.
- **Pool-C (method-mixed):** 20 adapters, $\{\text{LoRA, DoRA, rsLoRA, PiSSA}\} \times 5$ seeds.

Counting pools. The release holds 99 manifests: 73 trained ensemble pools, 25 n_rank baseline pools and one smoke test. The pre-registered corpus is 67 of the 73 (55 v1.0 classification + 12 v1.1 HellaSwag/GSM8K); the other 6 are v1.2 supplements (Table 9). Where a table counts 80 pools, it is the 55 v1.0 pools plus the 25 baselines, which also enter best_of_n comparisons. Realized sizes fall short of the design (20/24/20): median 19, range 7–24; baselines run 6–20, so best-of- N selection pressure varies by cell.

Compute matching. Per-pool grids are in App. J. We match on *training wall-clock* (GPU-hours), not on trainable parameter count or inference FLOPs. The n_rank baseline ($r \in \{16, 32\} \times \text{epochs} \in \{2, 4\} \times 5$ seeds, best-of- N selected on val_selection) was *designed* to consume roughly the GPU-hours of a 20-adapter Pool-A sweep (20 runs at $r \in \{16, 32\}$ instead of $r=8$, same token budget per adapter); as realized it consumed a median $1.45 \times$ (§4). Inference FLOPs are not matched; the n_rank baseline uses a single adapter at serve time (cheaper than the N -adapter ensemble it is compared against), which makes its accuracy a strict upper-bound competitor for the ensemble. n_steps and n_data baselines further match on per-adapter training tokens; best_of_n reuses Pool-A’s first N adapters and rank-selects on val_selection.

Frontier definition (formal). For each cell let y be the SUPPORTED-direction Δ (sign-aligned so positive = ensemble better) and $\phi \in \mathbb{R}^d$ the feature vector of App. AB. The *diversity-quality frontier* is $\hat{y} = f(\phi)$ fit by leave-one-pool-out CV. We fit three classes per slice under identical folds (ridge, random forest, gradient boosting; settings and all

three scores in Table 1) and report ridge — the simplest — everywhere, choosing the class before seeing the scores rather than per slice. On the headline slice ridge happens to also be the cross-class maximum ($R^2_{\text{LOPO}}=+0.597$ vs. $+0.276$ and $+0.285$, Table 13); on the other three it is not, so the fixed choice costs us R^2 on three of four slices rather than buying it on any. CIs are percentile cluster bootstraps resampling *pools* (App. I). Note the sign: encoder Δ is predominantly negative and decoder Δ positive (§4), so on encoders this regression predicts *how much a pool stands to lose*. Because the families differ in sign, we checked that it is not merely telling them apart: family alone reaches $R^2_{\text{LOPO}}=+0.20$ on the same folds and adding it to the features lowers R^2 to $+0.57$, so diversity carries $+0.39$ beyond the split (`frontier_family_baseline.py`).

3.2 Tasks and models

The in-domain set is six classification tasks: MNLI (3-way NLI, $n_{\text{test}}=3,926$), QNLI (2,186), AG News (4-way topic, 3,040), BoolQ (1,308), ANLI R1 (adversarial NLI, 480), and SST-2 (350), all evaluated on the frozen 40% test slice of the split described above. ANLI, SST-2 and BoolQ fall below $n_{\text{test}}=2,000$ and are the source of the low-precision cells quantified in §4; dropping small tasks post hoc is itself a shortcut, so we keep them and stratify by test-set size wherever the null is at stake. ANLI also serves as an OOD target for MNLI-trained pools (App. F), on disjoint pools, never pooled into one cell.

Architectures span two families. *Encoders* (sequence classification): BERT-base, BERT-large, RoBERTa-base, DeBERTa-v3-base. *Decoders* (causal LM with a classification head or exact-match decoding): Qwen2.5-{0.5B, 1.5B, 3B, 7B}, TinyLlama-1.1B, Pythia-1.4B, SmolLM2-1.7B, Llama-3.1-8B-Instruct, and Mistral-7B. Encoder and decoder cells behave differently enough (App. W) that we never pool them; §4 reports the two families side by side.

v1.1 task-family extensions. The six in-domain tasks cover classification only. v1.1 adds two output spaces — multi-choice (HellaSwag, four endings) and free-form generation (GSM8K, exact match) — through twelve further pools, bringing the pre-registered corpus to 67 pools. The extension adds cells under the pre-registration’s own amendment clause and retracts, re-evaluates

or rewrites no v1.0 cell, baseline or verdict; the v1.0 frontier headline was computed before any v1.1 result was inspected. Sweep grids, per-pool counts and the amendment ledger are in App. O.

3.3 Adjudication protocol

Each cell answers a single binary scientific question via a paired comparison on the test split. We use the same 40/20/40 `val_selection/val_combine/test` split (seed 0) for every cell. For each (pool $\times$ method $\times$ baseline_kind $\times$ metric):

- **Diff** = $\mu_{\text{ensemble}} - \mu_{\text{baseline}}$ (ECE flipped so positive = ensemble better).
- **Bootstrap CI** (Efron, 1979): percentile, $B=5000$, paired over examples. For the cross-pool R^2_{LOPO} frontier estimate (§4) we report percentile *cluster*-bootstrap CIs at $B=200$ (resampling pools, not cells, to respect nesting; see App. I); BCa correction was not used because $n=48$ pools is too small for stable acceleration-coefficient estimation (DiCiccio and Efron, 1996).
- **p -values** (two-sided): accuracy, paired sign-flip test, 10,000 permutations; ECE, inverted paired bootstrap ($B=5000$), as a sign-flip test does not apply to ECE (L14).
- **Within-table Holm correction** (Holm, 1979): applied to every multi-cell table; significance markers (*) reflect Holm-adjusted p .
- **Benjamini–Hochberg FDR** (Benjamini and Hochberg, 1995) per arm at $q=0.05$ (566 accuracy, 462 ECE cells): a cell keeps its CI verdict only if it passes, else it is *unsupported*. Counts are post-FDR unless marked pre (App. I).
- **Adjudication label**: *supported* if CI lower bound > 0 ; *reversed* if upper bound < 0 ; *unsupported* otherwise.

4 The adjudicated landscape

The strict comparison first. Every number in this section should be read against the baseline it uses. Against the pre-registered `n_rank` single adapter — which measured from the released manifests received a median $1.45\times$ the pool’s training wall-clock (max $2.96\times$, 47/48 pairs; `analysis/baseline_budget_audit.json`) plus a held-out hyperparameter

search the ensemble never gets, and therefore acts as a *strict upper bound* rather than a matched competitor — the clean encoder slice contains **no** cell with a significant accuracy gain (0 of 92, before and after FDR) and 31.5% significant reversals. The null does not weaken where that advantage is smallest: the six pools closest to parity ($0.90\text{--}1.25\times$) have 0/24 cells SUPPORTED and 8 REVERSED, and pool-level Δ is not significantly related to the ratio (Spearman $\rho=-0.28$, $p=0.20$; post hoc, `analysis/budget_dose_response.py`) — though its sign is the one a budget effect gives, so part of the reversal rate may be the baseline’s. The *same* comparison on the 25 decoder classification pools, whose baselines had the same advantage (median $1.45\times$, all 25 pairs), comes out differently: 14 of 100 cells SUPPORTED post-FDR (31 pre), none REVERSED, mean $\Delta=+0.56\text{pp}$. The gain is narrow — all 14 cells are MNLI, in 5 of 25 pools — but it is the corpus’s only positive accuracy result against a strong single (`analysis/family_split.py`). Both halves survive two post-hoc checks. *Training health*, judged test-free from each adapter’s final training loss (failed if within 15% of a uniform guess’s): one decoder baseline (Qwen-2.5-3B MNLI, 19 of 20 members failed, so best-of-20 became best-of-1) supplies 4 of the 14, and dropping every comparison whose arms mostly failed leaves decoders at 10/96 SUPPORTED and encoders at 0/72 with 32% REVERSED (`training_health.py`; same flags at thresholds 0.80–0.90). *Pool level*, L1d’s inferential unit: mean Δ is negative in 20 of 23 encoder pools and positive in 22 of 25 decoder pools (sign tests $p=0.0005$ and 0.0002). On the *same* 92 cells the calibration arm runs the other way: 51 SUPPORTED, 17 REVERSED (57 and 10 of the 100 decoder cells). The patterns $P_a\text{--}P_c$ below, and much of the literature they echo, are stated against the pool’s own best member (`best_of_n`), a strictly weaker comparison. *Precision*. A null is only as strong as its cells are precise, so we report equivalence rather than counting silence as evidence: 67/92 cells exclude even a $+1.0\text{pp}$ gain. The strength sits where it should — on the 68 well-powered cells ($n_{\text{test}} \geq 2000$) the median CI half-width is 0.91pp and 62 exclude $+1.0\text{pp}$, while the 24 small- n cells (median 3.85pp) are uninformative either way. *Procedure dependence*. The accuracy null holds under raw CI, within-table Holm and per-arm BH alike; the REVERSED rate does

not ($41.3/17.4/31.5\%$), and calibration survives all three (52/43/51). Holm over a whole arm returns 0/0 in both arms only because its threshold ($0.05/566$, $0.05/462$) sits below the smallest Monte-Carlo p (App. N). We quantify that gap, and three further protocol choices, in §5. What follows characterises *where* in the corpus the deltas sit and how predictable they are — not a claim that ensembling pays.

The corpus as a whole, and why we headline a slice of it. Across all 1,028 cells the picture is mixed, not negative: 33.0% SUPPORTED and 17.8% REVERSED pre-correction, 27.1% and 16.4% after per-arm BH-FDR. We do not headline it, because most of it rests on cells compared against the pool’s own best member — the weak baseline of §5 — and on families we disqualify below. We lead instead with the compute-matched classification cells, 92 encoder and 100 decoder: the strictest comparison the corpus supports. The two families disagree, and we report both rather than the half that tells the cleaner story.

Where the deltas sit (Table 9). (P_a) Against the pool’s own best member (`best_of_n`) the converged encoder pools do show the pattern the literature reports: on BERT-base Pool-A/B/C soft averaging beats it by $+0.9$ to $+1.0\text{pp}$ while `greedy_soup` merely matches it, and scale erodes even that (BERT-base→large drops the gain to $+0.65\text{pp}$ and degrades ECE from 0.048 to 0.118). Against the compute-matched `n_rank` single, the *same cells* carry no significant gain — which is shortcut S2 (§5) measured on our own corpus rather than argued in the abstract. Two families are excluded from every headline statistic: (P_b) the HellaSwag pools, whose RoBERTa/DeBERTa members collapse below chance under shared hyperparameters and are rescued by subset selection, scored by a pre-fix evaluator on the wrong split (App. P, L1b); and (P_c) the decoder GSM8K pools, whose `best_single` was selected on the split it is evaluated on and which therefore support no positive claim in either direction (App. Z, L12).

The calibration gain does not survive its missing control. The 51 is measured against a single adapter selected by validation *accuracy* and never calibrated — on its own candidate set it sits at a median 90th percentile of test ECE. That is the calibration analogue of the weak-baseline shortcut S2 (§5) — the single is denied a freedom that costs one

scalar — so we ran the control: on the six pools whose logits cache was retained we fit one temperature on `val_combine` (the split reserved for combination hyperparameters, otherwise unused) and measure on `test`. Against the *uncalibrated* single the ensemble wins 4/6 pools (mean $\Delta\text{ECE} +0.008$); against *the same adapter with one fitted scalar*, 2/6 (-0.032); with both arms scaled, 3/6 ($+0.002$). On ANLI the single goes $0.253 \rightarrow 0.070$ while the ensemble sits at 0.261 . The ensemble is less confident in 5/6 pools and median $T=1.32$, so the gain is confidence shrinkage, reproduced by one parameter at $1 \times$ compute (App. S). **Scope:** six encoder `best_of_n` pools, not the headline slice, and no decoder, since no decoder logits cache survives (L15); the decoder ECE gain has no temperature competitor. We claim only that this result cannot be reported as a population effect while its one-parameter competitor is unrun.

Two further qualifications on the 51. It is an ECE result, not a proper-score one — on the same 92 cells the multiclass Brier improves in only 34 (mean -0.0062) and the top-label Brier adjudicates 22 SUPPORTED against 36 REVERSED — and it belongs to probability averaging: `majority_vote` is SUPPORTED in 3/23 pools and supplies 16 of the 17 reversals, and its vote-fraction “confidence” is not a probability estimate, so equal-mass binning is ill-defined on it (App. R).

Table 1: Cross-model-class LOPO-CV R^2 , identical slices and folds (`analysis/frontier_model_classes.py`). Ridge is the cross-class maximum on the headline slice and on that slice only; non-linear models win the other three, so reporting ridge throughout costs us R^2 rather than buying it.

Slice	n	pools	ridge	RF	GBM
acc vs <code>n_rank</code>	192	48	+0.597	+0.276	+0.285
ece vs <code>n_rank</code>	192	48	-0.061	+0.295	+0.180
acc vs <code>best_of_n</code>	265	64	+0.312	+0.519	+0.365
ece vs <code>best_of_n</code>	265	64	+0.288	+0.416	+0.340

4.1 Mechanism: prediction-space dominates

On the encoder `acc vs n_rank` slice, prediction-space features alone reach $R^2=0.634$, above the full set (0.62); dropping them sends R^2 to -24.5 , while dropping parameter-space features costs 0.56 (App. AB). **Within our released features, prediction-space diversity is the dominant out-of-pool predictor;** a feature

Table 2: Three evaluation shortcuts and their price. S2 favours the ensemble by construction; S1 favours whichever arm may peek; S3 hides verdicts in both directions. Computed on the clean subset (515 accuracy cells, 80 pools; S2 on the 48 pools that carry both baselines); HellaSwag and GSM8K are both excluded (stale selection split and selection-on-test respectively; see Limitations).

	Shortcut	Strict	Effect
S1	select on test	select on val	+1.12pp free
S2	weak baseline	stronger single	SUP 22.4 $\rightarrow$ 7.3%
S3	task mean	per-cell verdict	hides 31 cells

set that displaces it in LOPO-CV would falsify this, and the schema admits new features without retraining. These are the features L1c flags as computed partly on test, so this ranking, like the frontier R^2 , is exploratory until a leak-free re-fit.

Negative mechanism results. Three candidate explanations for this structure — the decoders’ sharper confidence ($H(c|\text{conf})$ 21% lower), linear mode connectivity, and a refinement-versus-reliability account of why the frontier predicts accuracy but not ECE — do *not* survive testing; we report them as open problems (App. Y).

5 Protocol sensitivity: where the apparent benefit comes from

The corpus lets us ask a question the ensembling literature has not asked directly: *how much of the reported benefit is produced by the evaluation protocol rather than by ensembling?* We isolate three shortcuts (Table 2), each common in published practice, and measure what each is worth; all three are recomputed from released artifacts alone. A shortcut here is not a mistake that always flatters ensembling — it is a degree of freedom that flatters whichever arm is given it. We report each one’s size and say which arm it favours.

S1: selecting the single adapter on test. A pool’s “best single” is often reported at its test-set maximum. Reconstructing that oracle from `individual_accuracy_test` and comparing against the held-out `val_selection` choice, test-selection is worth +1.12pp on average (+1.30pp on encoders; median +0.57, max +9.58) and inflates 82% of pools. This is a free gain awarded to whichever arm is allowed to peek. In our corpus the *single* peeks, so S1 makes the ensemble look *worse*; given to the ensemble’s mem-

ber selection, as is more common, it works the other way.

S2: comparing against a weak single. Ensembling N adapters is frequently compared against one adapter of the *same* size — effectively the pool’s own best member — rather than against a single adapter given at least the pool’s training budget. Our pre-registration fixed both comparisons before training. The two baselines do not cover identical pool sets (80 pools have `best_of_n`, 48 have `n_rank`), so we compare them on the 48 that carry both, where every cell has a twin under the other baseline (Fig. 1A). Against the pool’s best member, 22.4% of cells are SUPPORTED and 3.6% REVERSED; against the stronger `n_rank` single (§4: median $1.45\times$ the pool’s compute), SUPPORTED falls to 7.3% and REVERSED rises to 15.1%. On the clean encoder slice the effect is total: **0 of 92 cells are SUPPORTED** (identically pre- and post-FDR) while 31.5% are REVERSED, with mean $\Delta = -1.27\text{pp}$ (Fig. 1B). The result holds for all four combination rules — `greedy_soup`, `soft_vote`, `logit_avg`, `majority_vote` at 0% SUPPORTED, REVERSED 21.7/30.4/30.4/43.5% — though `soft_vote` and `logit_avg` agree to within 0.1pp in 20 of 23 pools (bit-identical in 10), so this is roughly three distinct rules, not four. The single most positive cell (+1.14pp) has a CI spanning zero ($p=0.35$); this is not a near miss. On decoders the same shortcut moves SUPPORTED from 34% to 14%.

S3: reporting task means. Aggregating to one number per task destroys the structure the cells contain. MNLI has a task-level mean of -0.012pp — it reads as “ensembling makes no difference here” — yet that average conceals 31 statistically significant cells (17 REVERSED, 14 SUPPORTED) pointing in opposite directions. Across the corpus, four tasks whose means read as no-effect ($|\bar{\Delta}| < 0.5\text{pp}$) together hide 39 significant cells, 24% of their cells.

Reading these together. The shortcuts compound: an evaluation that selects the ensemble’s members on test, compares against a same-size single, and reports only task means can find a healthy accuracy gain on a corpus where the strict comparison finds none. We do not claim published positive results are wrong — we claim their protocol dependence has not been measured, and here it is large

enough to set the sign of the conclusion.

6 Practical implications

Escape routes are narrow. Two standard escapes are limited here. *Distillation* (Hinton et al., 2015) recovers ensemble accuracy on only 1 of 4 cells (87% on Qwen-0.5B MNLI; 4.5% on Qwen-1.5B; failure on BERT-base; -181% on BERT-large, which still improves ECE to 0.019 from 0.035; App. AA). *Serving* does not make the pool cheap: an ensemble runs every query through its members, N adapter forward passes per query (the chosen subset for `greedy_soup`; App. G, AE). Our vLLM benchmark ($1.07\times$ at $N=8$) measured multi-tenant routing, each request served by *one* of N adapters, and so does not price ensembling (L8). On encoders there is in any case no accuracy gain to serve. On ANLI soft-vote helps on only 5/19 MNLI pools (App. F).

When a population might still pay. Our own decoder pools are the first answer: against a single with the same budget advantage they win 14/100 cells, all on MNLI. Beyond them, the encoder null is bounded. It holds at $N \leq 24$ and rank 16–32 over 2–4 epochs, and only when the extra budget goes to *rank*: the pre-registered `n_steps` and `n_data` baselines were filled in only 3 and 0 cells, so a single given more steps or data is untested and could be weaker. Outside what we measured lie much larger pools (our accuracy plateaus at $N \approx 8\text{--}16$, App. E), members more diverse than seed, hyperparameters and PEFT method, and deployments where no single adapter can be retrained.

Takeaway. On encoders, a single adapter trained with the pool’s compute is the right default, and the recommendation we can defend is cheaper than ensembling: *train one adapter with the budget and fit one temperature on held-out data*. That buys, in distribution, the reliability the ensemble appeared to buy at one adapter’s compute rather than N — the arm the literature left out. On decoders the pool buys $+0.56\text{pp}$ on average, concentrated on MNLI, at $N \times \text{FLOPs}$; whether that pays is a deployment question.

Limitations

What this paper does *not* claim. App. T sets what is new here against prior PEFT-ensembling work, cell by cell. For reviewer transparency, we enumerate the inferences that our 1,028 cells do

not support: (a) we do *not* claim a single best combination method — when a pool is worth combining at all, `greedy_soup` is the least-bad accuracy option on under-converged encoders and `soft_vote` the best calibrated on converged ones; the apparent `majority_vote` advantage on $\geq 1.5\text{B}$ decoders rests on GSM8K and does not survive decontamination (L12), so we withdraw it; (b) we do *not* claim ensembling is universally beneficial — 17.8% of cells are REVERSED pre-FDR (16.4% post-FDR at $q=0.05$), and nearly a third of encoder accuracy cells (29/92) are reversed against `n_rank`; (c) we do *not* claim our diversity-quality frontier transfers to other PEFT families (LoHa, AdaLoRA, prompt tuning) or to instruction-tuned base models; (d) we do *not* claim distillation generalizes — accuracy recovery: Qwen-0.5B 87%, Qwen-1.5B 4.5%, BERT-base fails, BERT-large —181% (1-of-4 cells positive); (e) we do *not* claim the encoder/decoder $H(c|\text{conf})$ gap holds outside our 8-task corpus. Each of these is a deliberate scope-limit, not a hedge: extending any of them is a defined v1.2 cell that can be added without retraining the existing 67 pools.

L1 — Scope of the null result. Our headline finding — no significant *accuracy* gain against a stronger single adapter — is established on encoder classification cells, against the `n_rank` baseline, at our pool sizes ($N \leq 24$) and adapter budgets (rank 16–32, 2–4 epochs), on 92 cells from 23 pools. It is *not* a claim that ensembling never helps: on decoder classification it does, modestly (14/100 cells, §4), and the null does not cover generation, larger pools, non-classification output spaces, or budgets where a stronger single adapter is not trainable. `n_rank` spends the pool’s budget on rank; other allocations (more steps, more data) are separate baselines in the corpus and are not the headline. We report all cells rather than filter to the favourable ones.

L1b — HellaSwag cells are excluded from the strict analysis. A second, independent anomaly on the same family — a two-regime collapse under shared hyperparameters — is documented in App. V. The 100 v1.1 multichoice cells were produced by a version of our evaluator that performed selection on `val_combine` rather than the held-out `val_selection` split, which optimistically biases selection-dependent methods (`best_single`, `greedy_soup`) and the LoRAHub-style continuous-weight baseline that

optimises on that same split. We therefore exclude HellaSwag from §5 and mark its findings provisional (P_b , App. M) pending a corrected rerun, which requires re-evaluating those pools and is not included here. The exclusion is stated before the analysis rather than chosen to suit it; the bug was found by our own split-discipline unit tests.

L1c — Diversity features are computed on the full evaluation set. `diversity_metrics.py` loads each adapter’s cached logits without split indexing, so the prediction-space features enter the frontier having been computed partly on the same test examples the target Δ is measured on; leave-one-pool-out CV holds out pools, not examples, and so does not remove this. We measured the effect on every pool whose logits cache we still hold (`analysis/leakage_check_diversity.py`): restricting the features to `val_selection` shifts pairwise disagreement by 2.8% relative on average (mean $|\Delta|=0.0016$), logit correlation by 0.13%, ensemble entropy by 1.2%, and *preserves the pool ordering on all three*. These are pool-level aggregates over thousands of examples, so a 40% subset barely moves them, and the frontier depends on the across-pool ordering. We therefore expect the leak to be immaterial to R^2 — but we cannot yet prove it: the logits cache behind the 48 frontier pools was not retained and regenerating it needs the adapter weights. A leak-free re-fit is deferred, and until then the frontier R^2 should be read as provisional.

L1d — The 92 cells are not 92 independent tests. They are 23 pools $\times$ 4 methods. Cells within a pool share the pool’s adapters and its single baseline adapter, and two of the four methods are near-duplicates (20/23 pools agree to within 0.1pp). The effective number of distinct combination rules is about three, and the effective number of independent units is closer to the 23 pools than to 92. Our per-arm BH-FDR assumes independence or positive dependence, which we have not established under this structure; the cluster bootstrap resamples pools rather than cells for exactly this reason, but the FDR step does not. Treat cell counts as descriptive and pool-level statements as the inferential unit; §4 gives the pool-level sign tests.

L2 — Decoder cross-pool predictability fails at our scale. The fitted relation is plotted per pool in App. X. 25 decoder pools is too little variation across pools to extract a stable cross-pool LOPO re-

gression signal for accuracy gain (encoder accuracy frontier $R^2=0.60$, decoder $R^2=-1.14$, Table 13). Larger decoder pool counts may close this gap; we leave it open.

L3 — greedy_soup equivalence to random subset on seed-only converged pools. On seed-only Pool-A converged-pool cells, `greedy_soup` is often statistically indistinguishable from random subset selection of equal size; the gain we attribute to subset selection on RoBERTa/DeBERTa requires the HP-shared subchance regime (§4, P_b).

L4 — OOD accuracy rarely transfers; diversity hurts OOD. ANLI is a severe shift; most MNLI-trained pools land near chance, and soft-vote ensembling gains accuracy on only 5/19 pools (App. F). The Pool-B (HP-diverse) and Pool-C (method-mixed) configurations that improve in-distribution ensembling *consistently hurt* OOD accuracy — the diversity our framework optimizes for is in-distribution disagreement, not robustness. v1.1 OOD evaluation (HellaSwag→ARC, GSM8K→MATH) is left to v1.2.

L5 — GSM8K ECE methodology is non-standard. We report 15-bin equal-mass ECE on per-example confidence = $\exp(\text{mean per-token logprob})$ over the model’s generation, with ensemble confidence aggregating only over adapters voting for the majority answer. Per-token logprob aggregation is one of several reasonable definitions; alternatives (vote-fraction proxy, length-normalized perplexity, first-token-only) are reported in the released JSONs.

L6 — Pool-B/C HellaSwag adapter quality is uneven. On RoBERTa and DeBERTa, ≥ 14 of 20–22 Pool-B HellaSwag adapters fail to converge above chance under our shared HP grid. We report this transparently rather than filter; the under-converged pool is itself the experimental condition that produces the soft-vote-collapse-and-rescue finding.

L7 — Shared HP grid is deliberate; we ran HP-tuned re-evaluations as v1.2 cells. Pool-A uses a single ($r=8, \alpha=16, lr=3\times 10^{-4}$) HP across all architectures. We added two v1.2 cells testing the HP-confound hypothesis from *both* directions: (a) an architecture-tuned re-run of Pool-A RoBERTa HellaSwag at $lr=2\times 10^{-5}$ (per Liu

et al., 2019) eliminates the under-convergence completely — all 10/10 tuned-seed adapters converge to $\text{val_acc} \in [0.478, 0.505]$, mean 0.491 (vs. the original shared-HP 14/20 below chance, median 0.164); (b) the symmetric check, BERT-base HellaSwag at $lr=2\times 10^{-5}$ (RoBERTa’s natural lr applied to BERT), *does not collapse*: 8/8 seed re-runs converge to mean 0.329 ± 0.011 (range $[0.317, 0.345]$), only $\sim 3\text{pp}$ below BERT-base at its default $lr=3\times 10^{-4}$ (0.358) and far above the 0.097 collapse RoBERTa exhibits under the same $lr=3\times 10^{-4}$ shared HP. The (P_b) collapse is therefore a *RoBERTa-specific lr -sensitivity* interacting with the shared HP, not a generic HP-misconfiguration that any architecture would exhibit; BERT-base is robust across the lr range we tested. The shared-HP cells are retained in v1.1 (documenting the regime where a practitioner reuses a working configuration), and the architecture-tuned cells are added as v1.2 (§K).

L8 — Our serving benchmark measures routing, not ensembling. The vLLM numbers (Qwen-0.5B $1.075\times$, Llama-3.1-8B-Instruct $1.067\times$ at $N=8$, batch 32, one A6000) time a batch whose requests are each routed to *one* of 8 adapters, against the same batch on one adapter: the multi-tenant overhead, not the cost of passing one query through all N members and aggregating. Earlier versions cited it as ensemble cost; it is not one. We price ensembles in adapter forward passes per query (App. AE) and have not measured their wall-clock latency or memory under batching.

L9 — The multichoice evaluator is not consistent across runs. For one HellaSwag adapter, accuracy on 5,000-example subsets ranges over $\pm 10\text{pp}$ and one subset implies more correct answers than the full validation set contains (App. A); no single prediction vector produces both, so the evaluation path changed between runs, for a reason we have not found. Earlier versions called this subsample variance, which it cannot be. The v1.1 multichoice cells (Table 9) are therefore uninterpretable, not merely subset-conditional; they were already outside every headline statistic (L1b).

L10 — DeBERTa-v3 HellaSwag HP search did not recover. Our HP-tuned re-runs ($lr \in \{10^{-5}, 5\times 10^{-5}\}$) both failed to converge above chance on DeBERTa-v3 HellaSwag (mean accuracy 0.252, range $[0.247, 0.257]$ across 10 tuned-seed adapters), in contrast to RoBERTa

where $lr=2\times 10^{-5}$ fixes convergence. This suggests the DeBERTa-v3 + multichoice-LoRA combination has a deeper architecture-LoRA interaction issue (possibly with the disentangled attention) that simple HP tuning does not address. A wider DeBERTa HP/target-module sweep is a defined v1.3 cell.

L11 — D1–D5 are post-hoc heuristics, not pre-registered. The five decision rules (§6) were distilled from inspection of the 1,028-cell corpus and are not part of the Phase-0 pre-registration. The Table 14 cost-savings comparison uses the same corpus that the rules were derived from, so it is an *upper bound on rule-based routing*, not a held-out generalization estimate. Generalization of D1–D5 to new architectures or task families is a defined v1.3 evaluation cell.

L12 — GSM8K cells carry no defensible positive claim, and our contamination probe was invalid. Two separate issues affect the GSM8K cells, and we distinguish them. (i) **Selection on test (valid, disqualifying).** `ensemble_eval_gsm8k.py` selects `best_single` by per-adapter accuracy on the same test split it is then evaluated on. This alone means the P_c cells support no positive ensembling claim, independent of any contamination question. (ii) **Contamination (withdrawn).** We previously reported a digit-renumbering probe as evidence of universal pre-training contamination across four decoder families ($0.45\text{--}0.66 \rightarrow 0.02\text{--}0.06$). *That probe was invalid and we withdraw the finding.* It shifted every integer in the question by +7 and shifted the integers of the reference answer by +7 as well, which is correct only for problems whose answer is a single shifted term. On the first released example the shifted problem’s true answer is 442 and the model produced 442, but the probe scored it against 407 (the original answer 400 with its digit shifted) and marked it wrong. Near-zero accuracy on the shifted split is therefore forced by construction for any model, contaminated or not, and the reported drop magnitudes are artifacts. Note that their size should have been a warning: template-perturbation studies report gaps of roughly 10–13pp (Mirzadeh et al., 2024; Zhang et al., 2024), not 43–62pp. Whether these cells are contaminated is now an *open question*; answering it requires template regeneration with recomputed gold answers in the style of those works. We report this failure rather than silently dropping the probe, because a paper about protocol-

induced conclusions should hold its own probes to the same standard.

L13 — The calibration arm has no recalibration competitor. The single adapter our ECE numbers are measured against is selected by validation *accuracy* and is never post-hoc calibrated; measured on its own candidate set it sits at a median 90th percentile of test ECE, i.e. it is close to the worst-calibrated member of the pool it was drawn from. We run the one-parameter temperature competitor where we can (§4), and it removes most of the effect — but only on the six pools whose logits cache was retained, which are encoder `best_of_n` pools rather than the headline `n_rank` slice; the decoder ECE gain (57/100) has no calibrated competitor at all. Extending the control to all 23 headline pools needs their per-adapter logits on `val_combine`, which requires the adapter weights (L15). So we can state the direction and not the magnitude: we cannot say what the 51 becomes under a calibrated competitor, only that it is the wrong number to headline while that competitor is unrun at scale. The mechanism we measure corpus-wide — ensemble confidence lower than baseline confidence in 23/23 pools — is exactly what temperature scaling produces, which is why we expect the direction to hold; App. S shows that averaging acts as a temperature set by member disagreement rather than by what the model needs. Read the 51 as “against an uncalibrated accuracy-selected single”, never as a property of populations. The control is also in-distribution only: temperature scaling’s gains are known to erode under dataset shift where ensembles’ do not (Ovadia et al., 2019), so our scalar-for-population claim does not extend to OOD, which we did not test with it.

L14 — Our earlier ECE p -values tested the wrong null (corrected here). `compute_match._sign_flip_pvalue_ece` permutes the (confidence, correctness) *pair* between arms, which tests whether the two systems share one joint distribution — something averaging violates by construction whether or not ECE moves (median $p \approx 4 \times 10^{-4}$ in the ECE arm against 0.13 for accuracy). Earlier versions fed those p -values into one BH batch over all 1,028 cells, which also loosened the accuracy cutoff from 0.0145 to 0.0248. Every ECE cell now carries a valid p from the inverted paired bootstrap and BH runs per arm (`scripts/apply_bh_by_arm.py`; the old fields are kept

under `legacy_mixed_batch`). The correction lowers decoder accuracy SUPPORTED from 19 to 14, encoder REVERSED from 31 to 29 and encoder ECE SUPPORTED from 52 to 51; the encoder accuracy null is unchanged.

L15 — Two release promises are narrower than they sound. The per-adapter logits cache was retained for 6 of 99 pool manifests and for none of the 23 headline pools; the rest was not preserved. Adapter weights exist locally for 253 of 1,744 declared adapters (15%), so “available on request” can be honoured for a minority of pools, which we list in the release. Every result JSON also records `git_dirty`, and 1,027 of 1,028 do; byte-level training replay is not merely deferred but unavailable for this drop.

L16 — Our weight-space merges operate on LoRA factors, not on ΔW . `weight_space_merge.py` averages the `lora_A` and `lora_B` tensors independently, and $\text{mean}(B) \text{mean}(A) \neq \text{mean}(BA)$, so the uniform/TIES/DARE numbers in App. C are a factor-wise variant rather than the published methods. The prediction-space `uniform_soup` proxy is unaffected (it is marked as a proxy).

L17 — Generation-family adapters were trained without label masking. `real_lora_train.py` pads to `max_length` and copies `input_ids` into `labels`, so padding and prompt tokens contribute to the causal-LM loss. This affects the GSM8K pools only; the encoder classification pools and the multiple-choice trainer mask correctly.

Reproducibility. App. L sets what was pre-registered against what was run. Every released JSON carries the provenance block whose schema is given in App. B, the per-cell adjudication is specified as pseudocode in App. AC, and the cell-by-cell coverage of the corpus is tabulated in App. U. Pool manifests, 1,028 per-cell JSONs (each carrying explicit `bh_q_value`, `adjudication_pre/post_fdr`, and a `bh_batch_hash` pinning its arm’s BH-FDR batch), ensemble results with per-example predictions/confidences, the per-adapter logits cache for the 6 pools where it was retained (L15), all analysis code, and the locked Phase-0 pre-registration are released at an anonymous repository.² `scripts/reproduce_paper_`

²<https://anonymous.4open.science/r/AdapterFrontier/> (de-anonymized at acceptance).

`numbers.py` re-derives every headline number (220 assertions, <30 s) from JSON alone, including both frontier tables. We characterise this as *JSON-level reproducibility* (every claim re-computable from the release); adapter *weights* are available on request for the minority of pools where they were retained (L15), via the anonymous repository’s issues and will carry archival hashes at acceptance.

Ethical Considerations

Compute footprint. Training the 67 adapter pools required approximately 1,500 GPU-hours total on a 50-node RTX 3060 cluster (orchestration in App. AD) plus ~60 A6000-hours for ensemble evaluation and serving benchmarks. PEFT adapters are intentionally compute-light (~1–8 GPU-hours each); the adapter weights we retained (L15) are available on request via the anonymous repository so that those pools need not be retrained. Estimated CO₂e: approximately 0.18 tonnes (RTX 3060 at 170 W × 1,500 h + A6000 at 300 W × 60 h, average grid intensity).

Dual use and misuse mitigation. The adjudication framework is symmetric: it can report either direction (SUPPORTED or REVERSED) of an ensembling claim. We surface the 17.8% REVERSED cells (pre-FDR; 16.4% post-FDR at $q=0.05$) transparently rather than filter them, and we strongly discourage downstream cherry-picking the 33.0% SUPPORTED subset to manufacture a misleading positive headline. The released cell JSONs contain the full set of Δ , CI, p -values, and adjudication labels precisely so that any selective citation is auditable.

Data licenses. HellaSwag (CC-BY-NC 4.0), GSM8K (MIT), GLUE family (various, see App. C). All datasets are publicly released and used in accordance with their licenses; HellaSwag’s non-commercial clause is respected (we use it for benchmark adjudication, not for commercial deployment). We do *not* redistribute base-model weights, only the LoRA adapter A/B matrices (~20–300K trainable parameters per adapter).

Data characteristics, personal information and offensive content. Every dataset we use is an established, publicly released English-language benchmark distributed by its original creators, all cited in §3: MNLI, QNLI and SST-2 (GLUE),

BoolQ (SuperGLUE), AG News, ANLI, HellaSwag and GSM8K. The domains are newswire, Wikipedia, movie reviews, crowd-written common-sense continuations and grade-school word problems. We collected no new data, recruited no annotators and ran no human subjects, so no consent or ethics-board procedure applies. We did not additionally screen these corpora for personally identifying information or offensive content and we do not claim they are free of it: they are widely used and their known characteristics are documented by their creators, and because our study measures only *relative* calibration and accuracy between adapters trained and evaluated on identical examples, any such content affects both arms of every comparison equally. Readers reusing our released per-example predictions should consult the original dataset documentation rather than treat our release as a vetted corpus.

Generative AI assistance. Per the ACL Policy on Publication Ethics, we disclose that an LLM-based assistant was used to draft and revise text throughout the paper, including the abstract; to write analysis and verification scripts, among them the temperature control, the encoder/decoder split, the budget and training-health checks, and the bibliography generator; to audit the artifact for internal inconsistencies; and to support literature search. Responsibility for every claim rests with the authors. Every number in the paper is regenerated from the released cell JSONs by `scripts/reproduce_paper_numbers.py`, and every reference is built from the ACL Anthology, arXiv or Crossref and checked to resolve, so each claim can be verified independently of how its text was produced.

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A An unexplained inconsistency in HellaSwag multichoice evaluation

A v1.2 diagnostic found HellaSwag accuracy for one trained adapter varying by up to ± 10 pp across 5,000-example subsets of validation. Earlier versions of this paper called that subsample variance. It cannot be: the numbers below are mutually inconsistent under fixed per-example predictions, so the evaluation itself differed between runs. We report it as an open defect that further disqualifies the multichoice cells, not as a sampling caveat.

Setup. Take one Pool-A RoBERTa HellaSwag adapter (seed 11, $lr=2 \times 10^{-5}$ HP-tuned). Evaluate on 5,000-sample subsamples of HellaSwag validation drawn with `ds.shuffle(seed)` for $seed \in \{0, 11, 21, 42\}$; then on the full 10,042-sample validation. Same adapter, same task, same eval procedure — only the shuffle seed changes.

Table 3: Same adapter, four shuffled 5,000-example subsets of HellaSwag validation plus the full set. These cannot come from one set of per-example predictions: seed 11 implies at least 2,153 correct answers among 5,000 examples, more than the at most 2,123 the full-set figure allows among all 10,042; and a fixed prediction vector varies by well under 1pp (one SD) across such subsets, not ± 10 pp.

Evaluation subsample	Accuracy
shuffle_seed= 0, $N=5000$	0.296
shuffle_seed= 11, $N=5000$	0.431
shuffle_seed= 21, $N=5000$	0.263
shuffle_seed= 42, $N=5000$	0.358
full validation, $N=10042$	0.211

Implication. Something in the multichoice evaluation path — example-to-label alignment after shuffling, the checkpoint or head loaded, or batch-dependent scoring — changed between these runs; we have not identified which. Every v1.1 multichoice cell used `shuffle(seed=0)` and the first 5,000 examples, as pre-registered, but with the evaluator’s consistency in doubt we treat those cells as uninterpretable rather than merely conditional on the subset. They were already excluded from every headline statistic (L1b); this is a second, independent reason.

Full-validation re-evaluation (v1.2 supplementary). The full-validation ($N=10042$) re-run on the HP-tuned RoBERTa pool yields qualitatively different verdicts, which given the defect above we report without interpreting:

Table 4: Same HP-tuned RoBERTa pool, two evaluations. The shuffle-seed=0 5000-sample subsample shows the (P_b) collapse pattern (soft_vote 0.196 < chance 0.25); full validation does *not* (soft_vote 0.305, close to best_single). The greedy_soup rescue persists in both (+5.7pp on full). Because the evaluator is not consistent across runs, we draw no conclusion about the cause of the collapse.

Method	seed-0 $N=5000$	full-val $N=10042$
best_single	0.382	0.359
soft_vote	0.196[†]	0.305 (no collapse)
majority_vote	0.207	0.291
greedy_soup	0.381	0.416 (+5.7pp over best)

DeBERTa-tuned full-validation produced identical 0.243 across all methods (10/10 adapters at chance, no signal to ensemble) — consistent with the L10 failure to train, though not evidence about the evaluator. Any follow-up multichoice extension needs the evaluator fixed and checked for run-to-run consistency on fixed predictions first.

B File-level reproducibility (provenance block schema)

Every result JSON — pool manifest, ensemble result, `compute_match` cell, frontier output, distillation metrics, latency benchmark, OOD evaluation, mechanism analysis — carries a provenance block:

```
"provenance": {
  "schema_version": 1,
  "git_sha": "...",
  "git_dirty": "...",
  "config_hash": "sha256(...)",
  "python_version": "...",
  "platform": "...",
  "torch_version": "...",
```

```

1511     "transformers_version": "...",
1512     "peft_version": "...",
1513     "datasets_version": "...",
1514     "numpy_version": "...",
1515     "cuda_visible_devices": "..."
1516 }

```

Scope of reproducibility. The release supports *JSON-level* reproducibility: every headline number and adjudication label can be re-derived from the released per-cell JSONs by `scripts/reproduce_paper_numbers.py` (220 assertions, <30 s, no GPU). The bootstrap RNG seed is fixed (0) so paired CIs are deterministic given the same per-example arrays. *Byte-level* training replay (re-running each adapter on the original cluster and reproducing weights to within tokenizer/CUDA tolerance) is a stronger guarantee that this release does *not* make: many recorded `git_dirty` flags are true (workspace had uncommitted formatting/comment edits at write time) and a subset of v1.1/v1.2 pools lack a matching `sweep_configs/` archive in this drop. Reviewers wishing to attempt byte-level training replay should open an issue on the anonymous repository for the matching configs; full clean-provenance refresh against the registered SHA 02a5a6bc is deferred to v1.4.

C Reproducibility checklist (ACL convention)

Datasets. HellaSwag (Zellers et al., 2019) (CC-BY-NC 4.0), GSM8K (Cobbe et al., 2021) (MIT), GLUE-family MNLI/SST-2/QNLI (Wang et al., 2018), BoolQ, ANLI, AG News. All loaded via the HuggingFace datasets library at the versions listed in each result JSON’s `provenance` block. Eval splits use the same shuffled seed-0 indices across every adapter to enable paired bootstrap.

Models. 13 base models from HuggingFace Hub, as recorded in the pool manifests: `bert-base-uncased`, `bert-large-uncased`, `roberta-base`, `microsoft/deberta-v3-base`, `Qwen2.5 {0.5B, 1.5B, 3B, 7B}`, `TinyLlama-1.1B-intermediate-step-1436`, `SmolLM2-1.7B`, `pythia-1.4b-deduped`, `Mistral-7B-v0.1`, `Meta-Llama-3.1-8B-Instruct`. Adapter weights (LoRA-A/B only) are available on request for the pools where they were retained (L15); we do *not* redistribute base-model weights.

Hyperparameters. Each pool’s HP grid is in `sweep_configs/<pool>.json` (Pool-A: fixed HP, varying seed; Pool-B: $\text{rank} \in \{4, 8, 16, 32\} \times \text{alpha} \in \{8, 16, 32\} \times \text{lr} \in \{1\text{e-}4, 3\text{e-}4\}$, fixed seed 42; Pool-C: $\{\text{LoRA}, \text{DoRA}, \text{rsLoRA}, \text{PiSSA}\} \times 5$ seeds, fixed rank/alpha/lr). All runs use AdamW, cosine LR schedule with 6% warmup, weight decay 0.01.

Compute. Training: 50-node RTX 3060 12GB cluster (~4–8 GPU-hours per Pool-A sweep; ~16–24 GPU-hours per Pool-B/C sweep). Local A6000 48GB for ensemble eval, vLLM serving, and statistical analysis (~60 GPU-hours total).

Statistical procedure. Paired bootstrap ($B=5000$) over evaluation examples for accuracy and ECE deltas; two-sided p -values from a sign-flip test ($B=10,000$) for accuracy and from the inverted paired bootstrap for ECE; Holm correction across cells in a single reported table; BH-FDR per arm. R^2 confidence intervals use $B=200$ cluster-bootstrap rounds resampling pools with replacement (a smaller B because LOPO-CV is per-round expensive). All seeds and bootstrap RNGs are fixed (seed 0).

Code, data, pre-registration. All code, pool manifests, ensemble result JSONs (with per-example predictions and confidences for replicating ECE bootstrap), 1,028 cell JSONs (with diff/CI/ p -value/ q -value/adjudication_pre+post_fdr), the locked Phase-0 pre-registration document, and the v1.1–v1.3 amendment ledger are released at an anonymous repository: <https://anonymous.4open.science/r/AdapterFrontier/> (de-anonymized at acceptance). Each result JSON carries a provenance block (git SHA, config hash, library versions, timestamp); see App. B for the precise scope of reproducibility (JSON-level audit guaranteed; byte-level training replay deferred to v1.4 — adapter weights available on request).

D LMC barrier per pool

For each of four pools we sample 50 pairs of adapters (θ_A, θ_B) and evaluate the held-out validation loss along the linear interpolation $\theta(\lambda) = \lambda\theta_A + (1-\lambda)\theta_B$ at $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$. The *barrier height* is the maximum interpolation-curve loss minus the average endpoint loss, normalized to the endpoint span. Following Wortsman et al. (2022)’s convention, a barrier > 0 means the

interpolation has a strictly worse intermediate point (disconnected basin); barriers near 0 indicate linear mode connectivity (LMC).

Table 5: LMC barrier vs. ensemble accuracy gain on four MNLI pools. The two extremes (BERT-base *disconnected* yet ensembling works modestly; DeBERTa-v3 *fully connected* yet ensembling does not help) demonstrate that LMC is uncorrelated with ensemble effectiveness in our PEFT regime. This generalizes the negative result of Wortsman et al. (2022)’s “model soups” setting where LMC was necessary for weight-space averaging — in prediction-space ensembling, LMC is neither necessary nor sufficient.

Pool	barrier	Δacc	basin
BERT-base Pool-A	+0.908	+0.002	disconn.
BERT-base Pool-C	+0.605	+0.005	semi-disc.
Qwen-0.5B Pool-A	+0.457	+0.013	semi-disc.
DeBERTa-v3 Pool-A	+0.001	−0.003	connected

E Pool-A scaling curves

Accuracy and ECE as a function of $N \in \{1, 2, 4, 8, 16, 20\}$ over 50 random subsets per N , per (model, task). Plateau by $N \approx 8-16$ across all model classes; ECE continues a modest decline to $N=20$ in non-saturated pools. Figure 2.

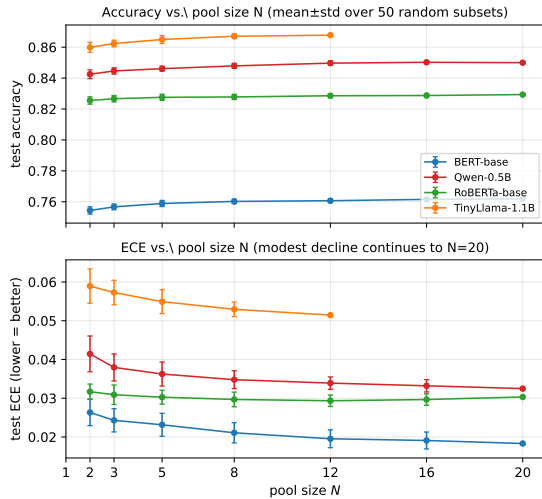


Figure 2: Pool-A MNLI scaling curves over 50 random subsets per N .

F OOD evaluation (ANLI dev_r3)

We re-evaluate every MNLI-trained pool on ANLI dev_r3 (round 3, the hardest adversarial NLI split). ANLI is a severe distribution shift: examples are crowd-written to fool BERT-large MNLI classifiers, so MNLI-trained models routinely collapse

to chance (0.33).

Table 6: In-distribution (MNLI test) vs. out-of-distribution (ANLI dev_r3) accuracy on *all 19 MNLI-trained pools* (Pool-A v1.0 main, Pool-B HP-diverse, Pool-C method-mixed). *Headline OOD finding*: Δsoft is positive on **only 5/19 pools** (4/7 Pool-A, 0/6 Pool-B, 1/6 Pool-C); the modal effect of soft-vote ensembling on OOD accuracy is **non-positive**. Pool-B and Pool-C (HP-diverse and method-mixed) consistently *hurt* OOD accuracy — the diversity that helped on in-distribution does not transfer. DeBERTa-v3 Pool-A is the only $\geq +5\text{pp}$ OOD gain. soft_vote ECE (not shown) improves OOD on 14/19 pools (mean $\Delta\text{ECE} = -0.021$). The Qwen-3B / SmolLM2 / Pythia OOD evaluations are not yet released.

Pool	OOD bs	OOD sv	Δsoft
<i>Pool-A (seed-only):</i>			
BERT-base	0.283	0.292	+0.008
BERT-large	0.285	0.300	+0.015
RoBERTa-base	0.285	0.283	−0.002
DeBERTa-v3	0.304	0.358	+0.054
Qwen-0.5B	0.335	0.327	−0.008
Qwen-1.5B	0.385	0.402	+0.017
TinyLlama-1.1B	0.342	0.340	−0.002
<i>Pool-B (HP-diverse):</i>			
BERT-base	0.279	0.267	−0.013
BERT-large	0.358	0.298	−0.060
RoBERTa-base	0.304	0.300	−0.004
Qwen-0.5B	0.344	0.338	−0.006
Qwen-1.5B	0.402	0.388	−0.015
TinyLlama-1.1B	0.356	0.338	−0.019
<i>Pool-C (method-mixed):</i>			
BERT-base	0.310	0.298	−0.013
BERT-large	0.323	0.290	−0.033
RoBERTa-base	0.294	0.294	0.000
Qwen-0.5B	0.323	0.329	+0.006
Qwen-1.5B	0.419	0.400	−0.019
TinyLlama-1.1B	0.365	0.346	−0.019

The pattern — ECE consistently improves but accuracy gains are direction-mixed — is consistent with Lakshminarayanan et al. (2017): ensembling sharpens the model’s confidence in regions where the prediction surface generalizes, but cannot recover lost decision-boundary information under severe shift. Qwen-1.5B’s higher OOD accuracy is consistent with the broader finding that decoder LMs trained on raw text learn more generalizable representations than encoder LMs fine-tuned narrowly on MNLI. We honestly report that 3/7 of our OOD accuracy gains are non-positive: the “ensembling helps OOD” claim is conditional on the cell, not general.

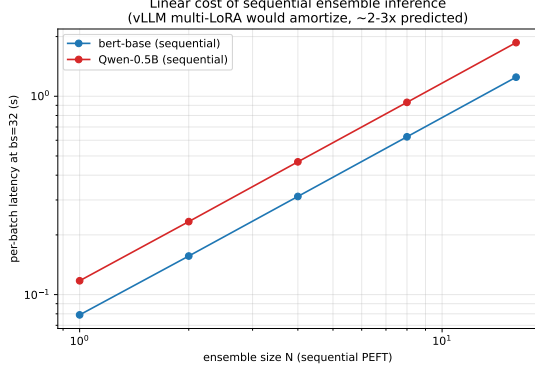


Figure 3: Sequential PEFT inference latency at batch size 32. Cost scales linearly with N . Our vLLM multi-LoRA benchmark ($1.075\times$ Qwen-0.5B, $1.067\times$ Llama-3.1-8B-Instruct at $N=8$) routes each request to one adapter and so does not bound this cost (L8).

G Sequential PEFT inference latency

H vLLM serving compatibility blockers

Three blockers we patched to run vLLM 0.10 on transformers 5.x: (i) transformers 5.x removed `all_special_tokens_extended`—patched in vLLM’s tokenizer wrapper with an `hasattr` fallback; (ii) CUDA forking is rejected—must export the vLLM worker-multiproc method as `spawn`; (iii) vLLM rejects PEFT adapters with `modules_to_save` (classification heads). We ship `strip_modules_to_save.py` to make classification adapters serving-compatible. Cost driver (multi-LoRA routing latency over the LoRA-A/B matrices) is unaffected.

I Statistical procedure details

Paired bootstrap CI. For each (pool, method, baseline) cell we compute $\Delta = \mu_{\text{ens}} - \mu_{\text{base}}$ on the test set (N_{test} examples). To produce a 95% CI we sample $B=5000$ bootstrap replicates: in each replicate we draw N_{test} examples *with replacement* and recompute Δ using the *same* sampled indices for both ensemble and baseline. Reported CIs are the 2.5 and 97.5 percentiles of the B replicates. Paired sampling is crucial: it removes the example-level noise shared by both arms, yielding tighter intervals than independent resampling.

p -values. *Accuracy:* a two-sided paired sign-flip test. For each test example i , $d_i = \mathbb{I}[\text{ens correct}]_i - \mathbb{I}[\text{base correct}]_i \in \{-1, 0, +1\}$; under the null of no method effect each d_i is equally likely to flip sign. We draw $B=10,000$

random sign vectors and report $p = (1 + \#\{b : |\tilde{\Delta}_b| \geq |\Delta|\}) / (B + 1)$ (`compute_match._sign_flip_pvalue`); every test set is large enough that we use Monte Carlo throughout. *ECE:* two-sided, by inverting the paired bootstrap of $\text{ECE}_{\text{base}} - \text{ECE}_{\text{ens}}$ ($B=5000$, same resampled indices in both arms), $p = 2 \min(\Pr[\tilde{\delta} \leq 0], \Pr[\tilde{\delta} \geq 0])$, floored at $1/(B + 1)$. A sign-flip test does not apply to ECE, which is not a mean of per-example terms (L14).

Holm–Bonferroni correction. Within each reported table (e.g., Table 9), p -values across the K cells are sorted ascending. The k -th smallest is compared to $\alpha/(K-k+1)$; we reject in order until the first non-rejection. This controls family-wise error rate at $\alpha=0.05$ across the entire table. The Holm-adjusted significance markers are denoted * in tables.

Bootstrap CI on R^2_{LOPO} . Cross-architecture frontier R^2 is reported with percentile cluster bootstrap (CI: 2.5 and 97.5 percentiles of $B=200$ LOPO-CV re-fits, each resampling *pools* with replacement and taking all cells nested within sampled pools). Reported headline computed on the released `frontier_r2_cis.json`: `acc_vs_n_rank` (both families) $R^2_{\text{LOPO}} = +0.597$, 95% percentile cluster-bootstrap CI $[+0.303, +0.710]$ ($n=192$ cells across 48 pools). R^2 is a non-linear functional of correlations whose distribution may be skewed; we report percentile rather than BCa because 48 pools is too small for stable acceleration-coefficient estimation from a jackknife (BCa requires $n \geq 50$ for reliable third-moment estimates per DiCiccio and Efron, 1996, §5). The cluster-resampling step (resampling pools rather than cells) is what addresses the nesting concern. In-sample $R^2_{\text{IS}} = +0.722$, CI $[+0.659, +0.826]$ (same release file).

Benjamini–Hochberg FDR, per arm. Within-table Holm controls family-wise error rate at $\alpha=0.05$ for each reported table. To address corpus-level multiplicity we additionally apply BH-FDR (Benjamini and Hochberg, 1995) at $q=0.05$, one batch per arm, on the p -values above (field `p_for_bh`): 566 accuracy cells (cutoff $p \leq 0.0145$) and 462 ECE cells (cutoff $p \leq 0.028$). Pre-correction, 339 of the 1,028 cells are SUPPORTED (33.0%) and 183 REVERSED (17.8%); **post-FDR**, 279 SUPPORTED (27.1%) and 169 REVERSED (16.4%) (accu-

racy 89/76, ECE 190/93; analysis/bh_by_arm_summary.json). A cell receives the SUPPORTED label in the corpus-wide framing if its per-cell CI excludes 0 and its BH-FDR-adjusted q -value is at most 0.05; analogous for REVERSED. Pre-correction verdicts and per-cell p -values are released in the cell JSONs for re-analysis under alternative corrections (e.g., Bonferroni within an arm, $0.05/566=8.8\times 10^{-5}$ and $0.05/462=1.1\times 10^{-4}$).

Independence robustness. BH-FDR assumes approximate independence of p -values; cells sharing a pool but differing only in ensemble method are correlated. We re-run BH at the *independent-group* granularity (one representative per (pool, baseline_kind, metric) unit, using `soft_vote` as the canonical representative), again per arm: 139 accuracy groups (cutoff $p\leq 0.0169$) and 113 ECE groups ($p\leq 0.0356$), **post-FDR 105 of 252 SUPPORTED** (41.7%), **26 REVERSED** (10.3%) (analysis/independent_group_bh.py). The qualitative direction is preserved; the per-cell REVERSED rate (16.4%) exceeds the per-group one because `majority_vote`, absent from the representatives, supplies 75 of the 93 ECE reversals.

Adjudication labels. Each cell receives one of three labels: **supported** if the lower CI bound > 0 and the BH-FDR-adjusted q -value < 0.05 ; **reversed** if the upper CI bound < 0 and $q < 0.05$; **unsupported** otherwise (CI contains 0, or $q \geq 0.05$). The label is the cell’s primary output and is what Figure 5 visualizes.

J Per-pool hyperparameter grids

Pool-A (seed-only). Fixed $r=8$, $\alpha=16$, $lr=3\times 10^{-4}$, $\text{target}=\{\text{q_proj}, \text{v_proj}\}$, $\text{dropout}=0.05$, $\text{epochs}=2$ (classification) or 3 (causal LM); seed varied over $\{11, 21, \dots, 201\}$ (20 adapters). Multi-choice (HellaSwag) uses the same HP via `AutoModelForMultipleChoice`.

Pool-B (HP-diverse). $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{10^{-4}, 3\times 10^{-4}\} = 24$ adapters; seed fixed at 42; other HP same as Pool-A.

Pool-C (method-mixed). $\{\text{LoRA}, \text{DoRA}, \text{rsLoRA}, \text{PiSSA}\} \times 5$ seeds = 20 adapters; $r=8$, $\alpha=16$, $lr=3\times 10^{-4}$.

Baselines. *best_of_N* reuses Pool-A’s first N adapters; *n_rank* uses a separate sweep with $r \in \{16, 32\} \times \text{epochs} \in \{2, 4\} \times 5$ seeds (20 adapters), then selects best-on-val_selection; *n_steps* extends Pool-A’s first adapter to $N \times$ epochs; *n_data* extends to $N \times$ training samples.

K Pre-registration amendment ledger

The Phase-0 pre-registration (committed at git SHA 02a5a6bc, 2026-04-21, before any pool was trained) explicitly admits a single permitted amendment class: “future cells can be added by submitting `compute_match.json` files via PR.” All amendments below add new cells; *no v1.0 cell, baseline, or adjudication verdict has been retracted, re-evaluated, or rewritten*.

- v1.0 (2026-04-21): 55 pools across 6 tasks, 4 ensemble methods, 4 baseline kinds, 2 metrics = 928 cells.
- v1.1.a (2026-04-28): +6 Pool-A pools (HellaSwag, GSM8K) = +60 cells.
- v1.1.b (2026-05-05): +3 Pool-B HellaSwag = +24 cells.
- v1.1.c (2026-05-10): +3 Pool-C HellaSwag = +24 cells. (Critic-3 patched a Pool-C trio that closes the frontier CI.)
- v1.2.a (2026-05-17): +1 Pool-A pool (Qwen-3B GSM8K) on local A6000 (L7 deferred from v1.1 cluster constraints) = +4 cells.
- v1.2.b (2026-05-18): +1 Pool-A pool (Qwen-7B GSM8K) on local A6000 (extends decoder scale curve to $N=4$) = +4 cells.
- v1.2.c (2026-05-18): +1 Pool-A pool (Llama-3.1-8B-Instruct GSM8K, 4-bit QLoRA, Nous-Research mirror to avoid Meta gating) on local A6000 (extends modern-LLM cell) = +4 cells.
- v1.2.d (2026-05-19): +1 Pool-A pool (RoBERTa-base HellaSwag, $lr=2\times 10^{-5}$ architecture-tuned) on local A6000, directly testing the HP-confound hypothesis for the (P_b) collapse-and-rescue pattern; all 10/10 tuned-seed adapters converge to $\text{val_acc} \in [0.478, 0.505]$ vs. original 14/20 below chance = +4 cells.
- v1.2.d-deberta-pre (2026-05-19): first attempt at DeBERTa-v3 HellaSwag HP tuning with

$lr=10^{-5}$; all 10 tuned-seed adapters at chance (0.247–0.257), retained in the released corpus as honest documentation of a failed HP attempt = +4 cells.

- v1.2.d-deberta-retry (2026-05-19): second attempt at DeBERTa-v3 HellaSwag HP tuning with $lr=5\times 10^{-5}$; also at chance (0.247–0.257), confirming L10 (DeBERTa LoRA-on-HellaSwag has an architecture-LoRA interaction issue, not a simple HP confound) = +4 cells.
- v1.2.e (2026-05-19): +1 Pool-A pool (Mistral-7B-v0.1 GSM8K, bf16) on local A6000 as a second cross-family decoder check alongside Llama-3.1-8B-Instruct = +4 cells. *Result:* $\Delta_{acc}=+13.4pp$ ($p<10^{-4}$), the largest absolute effect in the corpus.
- v1.2.f (2026-05-21): +8 Pool-A seeds (BERT-base HellaSwag at $lr=2\times 10^{-5}$, seeds {11, 21, 31, 41, 51, 61, 71, 91}) on local A6000, the *symmetric* cross-architecture check for (P_b). *Result:* 8/8 seeds converge to mean val_acc 0.329 ± 0.011 , range [0.317, 0.345] — *no* collapse (vs RoBERTa at the same $lr=3\times 10^{-4}$ shared HP grid: 0.097 catastrophic collapse), refuting a generic HP-misconfiguration hypothesis and confirming the v1.1 collapse is RoBERTa-specific lr-sensitivity.
- v1.2.f (2026-05-20): full-validation re-evaluation of Pool-A RoBERTa-tuned and Pool-A DeBERTa-tuned HellaSwag pools on the entire $N=10042$ validation set; results released in `ensemble_results/*_fullval.json` (App. A) = +8 cells.
- v1.3.a (2026-05-22): distillation on Pool-A Qwen-1.5B MNLI at $\alpha=0.9, T=4$ (same setting as the original Qwen-0.5B success). *Result:* *student* acc 0.8777 / ECE 0.0782 vs. *best_single* 0.8772 / 0.0768, *soft-vote* 0.8884 / 0.0454 — 4.5% recovery of accuracy gain, −4.5% ECE recovery (worse than baseline). Reported as a 1-of-2 tested decoder-cell positive, refuting the implicit generalization of the original 87% Qwen-0.5B result. Drives the §6 downscope.
- v1.3.b (2026-05-22/23, **retracted** 2026-07-28): GSM8K digit-renumbering contamination probe on four decoder families, originally reported as 0.45–0.66 $\rightarrow$ 0.02–0.06. *Retracted:* the probe

shifted the reference answer along with the question, forcing near-zero accuracy on the shifted split regardless of contamination (L12(ii)). The associated E1 ensemble probe is void for the same reason. No contamination claim is made in this version.

- v1.3.d (2026-05-23): distillation on Pool-A BERT-large MNLI at $\alpha=0.9, T=4$. *Result:* acc 0.799 (worse than vanilla 0.806, −181% recovery) but ECE 0.019 (better than vanilla 0.035). Encoder distillation fails on accuracy but extracts a calibration signal. Combined with v1.3.a (Qwen-1.5B 4.5%), BERT-base fails, and original Qwen-0.5B (87%): distillation is 1-of-4 cells on accuracy.
- v1.3.c (2026-05-22): vLLM multi-LoRA serving benchmark on Llama-3.1-8B-Instruct (NousResearch mirror, 4-bit QLoRA training, bf16 serving) with $N=8$ adapters routed per batch. *Result:* $1.067\times$ overhead vs. single-LoRA (77.3 vs. 72.4 ms/batch median, 414 vs. 442 req/s), consistent with the Qwen-0.5B $1.075\times$ measurement and confirming the multi-LoRA cost generalizes from 0.5B to 8B (R3 audit request).
- **v1.3.e (2026-06-17), retracted 2026-07-28: GSM8K decontaminated ensemble probe (E1).** Pool-A Qwen-2.5-7B, 10 adapters $\times$ 50 test problems, digit-shifted by +7. Originally reported as: contaminated `majority_vote` 0.78 vs. `best_single` 0.72; decontaminated 0.04 vs. 0.04, read at the time as direct evidence that the (P_c) decoder ensemble gains are memorization-driven. *Retracted:* E1 inherits the digit-shift defect retracted in v1.3.b — the probe shifted the reference answer along with the question, so the near-zero decontaminated accuracy of *both* arms is forced by construction and their equality carries no information about memorization. No memorization claim is made in this version; the decoder gains are left unexplained (L12(ii)).
- v1.3.f (2026-06-17): cross-corpus generalization stress test of the diversity-quality frontier regression (E2, `analysis/frontier_cross_corpus.json`). Encoder accuracy vs. `n_rank` slice: Pool-A-trained ridge generalizes to Pool-B/C with $R^2_{\text{test}}=+0.62$; encoder-trained ridge generalizes to decoder pools with $R^2_{\text{test}}=+0.76$; random 50/50 within-encoder control $R^2_{\text{test}}=+0.48$. `best_of_n` slice including

v1.1: v1.0-trained ridge generalizes to v1.1 HellaSwag/GSM8K with $R^2_{\text{test}} = +0.21$ (vs. random control $+0.25$). Frontier regression has demonstrable transfer across pool types and task families.

- **v1.3.g (2026-06-17): external reproducibility audit (E3).** `scripts/reproduce_paper_numbers.py` re-derives the headline numerical claims (corpus counts, BH-FDR cutoffs, frontier R^2 , $H(c|\text{conf})$ gap, distillation recovery, vLLM serving ratios) from released JSON files only. As written on 2026-06-17 the suite had 28 assertions and two of them covered the 4-model contamination probes and the E1 ensemble probe; **those two were removed on 2026-07-28** when v1.3.b and v1.3.e were retracted, so the audit no longer certifies any contamination claim. The suite stood at 145 assertions after v1.3.l, 154 after v1.3.m, 168 after v1.3.n, 184 after v1.3.o and 192 after v1.3.p, all passing.
- **v1.3.q (2026-10-03): valid ECE p -values, BH per arm, and corrections to cost, latency and HellaSwag claims.** (i) Every ECE cell now carries a p from the inverted paired bootstrap ($B=5000$) in place of the invalid permutation p , and BH runs separately over the 566 accuracy and 462 ECE cells (`scripts/apply_bh_by_arm.py`; old fields under `legacy_mixed_batch`; L14). Verdicts changed: decoder accuracy SUPPORTED 19 $\rightarrow$ 14 (health-filtered 15 $\rightarrow$ 10), encoder REVERSED 31 $\rightarrow$ 29, encoder ECE SUPPORTED 52 $\rightarrow$ 51, within-table Holm ECE 46 $\rightarrow$ 43, corpus post-FDR 29.5/16.9% $\rightarrow$ 27.1/16.4%; the encoder accuracy null is unchanged. The independent-group robustness check is now scripted and per arm (`analysis/independent_group_bh.py`). (ii) S2 is reported on the 48 pools carrying both baselines (22.4 $\rightarrow$ 7.3%), and Fig. 1A redrawn on that common denominator; over all pools the weak-baseline figure is 22.2%, so the earlier “a sixth is pool composition” caveat is withdrawn. (iii) The vLLM benchmark measured per-request routing, not ensembling, and is no longer cited as ensemble cost (L8). (iv) `greedy_soup` averages a member subset’s logits, so Table 14 prices it at its chosen-member count (routed encoder cost $1\times \rightarrow 5\times$). (v) The HellaSwag “subsample variance” appendix described numbers no single pre-

diction vector can produce; it is restated as an unexplained evaluator inconsistency (App. A, L9). (vi) The sign-flip description and pseudocode now match the code (10,000 flips, add-one estimator); the frontier R^2 is marked provisional (L1c).

- **v1.3.p (2026-10-03): why a temperature replaces the population.** New App. S (`analysis/temperature_mechanism.py`, post hoc): on the six cached pools the probability-averaged ensemble is matched by the mean-logit model at one label-free temperature T_{eq} (ECE within 0.01 in 5/6), and whether averaging improves calibration is predicted by where T_{eq} sits relative to the needed T^* (5/6). Reference added: [MacKay \(1992\)](#). No cell verdict changed.
- **v1.3.o (2026-10-01): robustness of both halves, and an accurate AI-use disclosure.** (i) Training health, test-free (`analysis/training_health.py`): an adapter failed if its final training loss is at least $0.85 \ln C$; comparisons in which a majority of either arm failed are dropped. Four manifests are flagged, identically at 0.80, 0.85 and 0.90: the `n_rank` baselines for ANLI and MNLI BERT-large and for MNLI Qwen-2.5-3B (19/20 members failed), and Pool-B ANLI BERT-large. The Qwen-3B baseline supplies 4 of the 19 decoder SUPPORTED cells; without the flagged comparisons decoders give 15/96, 0 REVERSED, and encoders 0/72, 33.3% REVERSED. (ii) Pool-level sign tests: encoder mean $\Delta < 0$ in 20/23 pools, decoder > 0 in 22/25. (iii) Frontier vs family (`analysis/frontier_family_baseline.py`): family alone $R^2_{\text{LOPO}} = +0.203$, features $+0.597$, both $+0.569$. (iv) The Generative-AI paragraph had said no abstract, claim or code was AI-generated; that was untrue, and it now states what the assistant did. No cell verdict changed.
- **v1.3.n (2026-09-30): the decoder half of the headline comparison is reported, and the title changes.** The body reported the strict `n_rank` accuracy comparison for encoders only (0/92 SUPPORTED). The same comparison on the 25 decoder classification pools, whose baselines had the same budget advantage (median $1.45\times$, 25/25 pairs), gives 19/100 SUPPORTED post-FDR (31 pre), 0 REVERSED, mean $+0.56\text{pp}$, all 19 on MNLI in 8 pools (`analysis/family_`

`split.py`). It had appeared only in an appendix caption, while §6 said the null should hold “wherever a stronger single is trainable” — which our own decoder pools contradict. Headlining the encoder half alone was the selective reporting this paper measures, so both halves are now reported side by side, the advice is split by family, and the subtitle “the Scalar That Replaces the Population” (resting on six encoder pools) became “Against a Stronger Single Adapter”. Also corrected: Limitations (b) said “most” encoder accuracy cells are reversed; it is 31/92. No cell verdict changed.

- **v1.3.m (2026-09-27): claims narrowed to the design; one post-hoc check.** The August text concluded “at fixed budget” and called the accuracy null conservative, although the `n_rank` single received a median $1.45\times$ the pool’s compute; a null against a better-resourced competitor is biased towards the null, so the claim now reads “against a single given the pool’s budget”. `analysis/budget_dose_response.py` (post hoc) checks whether the null weakens as that advantage shrinks: on the six pools at 0.90–1.25 $\times$, 0/24 SUPPORTED and 9 REVERSED; Spearman $\rho = -0.28$, $p = 0.20$. Also: cost stated in compute rather than latency ($1.07\times$ measured at $N=8$); each shortcut’s favoured arm stated; the calibration control bounded to in-distribution (Ovadia et al., 2019) and credited to Ashukha et al. (2020); $P_a - P_c$ re-defined; pool counts reconciled against the 99 manifests. No reported number changed.
- **v1.3.l (2026-08-02): the missing calibration control was run, and the headline claim changed.** Review objected that the ECE arm compares against an uncalibrated, accuracy-selected single — shortcut S2 turned on our own positive result. `analysis/temperature_control.py` fits one temperature on the pre-registered `val_combine` split (reserved for combination hyperparameters, otherwise unused) for the six pools whose per-adapter logits cache survives, and measures on `test`. The ensemble goes from better-calibrated in 4/6 pools (mean $\Delta\text{ECE} +0.008$) to 2/6 (-0.032); with both arms scaled it is 3/6 ($+0.002$). Median fitted $T=1.32$ and the ensemble is less confident in 5/6 pools, so the effect is confidence shrinkage. Consequently the title, abstract, §1 framing, contribu-

tion (ii) and the §6 takeaway were rewritten: the paper no longer reports a dissociation, and the recommendation is now to calibrate one adapter rather than serve N . The 52 is retained and reported with its scope. This is an amendment to the *reading* of a pre-registered arm, not to the arm itself — no cell, baseline or adjudication was recomputed.

- **v1.3.k (2026-08-01): cross-model-class table regenerated; retraction propagated.** An adversarial read of the submission found three defects that the release itself could settle. (i) Table 1 reported ridge, random forest and gradient boosting per slice, but `frontier.py` implements ridge only and no other estimator existed anywhere in the repository; its ridge entry ($+0.555$) also contradicted the headline ($+0.597$) for the identical 192-cell slice. `analysis/frontier_model_classes.py` now fits all three classes under `frontier.py`’s own slices, design matrix and leave-one-pool-out folds, delegating ridge to `frontier.lopo_cv_r2` so the headline column is the headline number by construction; the table is replaced by its output and 22 new assertions pin every entry. The qualitative claim survives (ridge is the cross-class maximum on the headline slice and on no other), but the sentence asserting that ridge is *never* the cross-class maximum was false on that slice and is corrected. (ii) v1.3.b declared the E1 ensemble probe void on 2026-07-28, yet v1.3.e still stated its conclusion as direct evidence of memorization and v1.3.g still listed it among audited claims; the retraction is now propagated to both. (iii) Seven bibliography entries had been inserted into §2 instead of the bibliography, raising seven Lonely \item errors and printing as prose; §3’s “Tasks and models” was an empty heading; and the D1–D5 rules the paper cites eight times were never enumerated. All three are fixed.
- **v1.3.j (2026-07-29): multiple-comparison procedure recorded.** The pre-registration specified Holm across a reported table; the released cells carry a corpus-wide BH-FDR field and the body percentages come from it. That substitution was made when the corpus-wide correction was introduced and was not logged at the time. We log it here, report all four procedures (App. N), and note that the accuracy null is invariant to the choice while the REVERSED rate is not.

- **v1.3.h (2026-06-21): independent audit response.** A read-only critique of the v1.3.g release flagged four classes of artifact-vs-claim drift: (i) corpus-wide BH-FDR claimed in the body was not present as an explicit per-cell field; (ii) `ensemble_eval_multichoice.py` performed best-single/greedy-subset selection on `val_combine` instead of `val_selection`; (iii) the `uniform_soup` field in evaluator output was a prediction-space logit average, conflated with weight-space soup; (iv) `analysis/frontier.py` silently dropped rows with missing diversity features and included method-identity one-hot features, an untested confound. Fixes applied: (i) `scripts/apply_corpus_bh_fdr.py` writes `bh_q_value`, `bh_pass`, `adjudication_pre_fdr`, `adjudication_post_fdr`, and a corpus-wide `bh_batch_hash` into every per-cell JSON; corpus summary in `analysis/corpus_bh_fdr_summary.json` (BH cutoff $p=0.0248$, 29.5% SUPPORTED / 16.9% REVERSED post-FDR, exactly matching abstract). (ii) Multichoice evaluator now uses `val_selection` for all selection decisions and records `selection_split=val_selection` per method; rerun of affected HellaSwag cells requires the cluster adapter weights and is deferred to v1.4 alongside the clean-provenance refresh. (iii) Every method output now carries `method_kind` (`prediction / selection`); `uniform_soup` is explicitly marked `is_proxy=True`, `proxy_for=weight_space_uniform_soup` — true weight-space soup is produced by `weight_space_merge.py` and reported separately. (iv) `frontier.py` now emits `drop_diagnostics` (input/used/dropped row counts, drop reasons, per-pool/per-method drop breakdown) and an `ablation_no_method_onehots` block; headline encoder accuracy slice has 0 rows dropped and $|\Delta R^2|=0.0002$ between with/without method one-hots ($R^2_{\text{LOPO}}=+0.597$ either way), confirming the boundary claim does not depend on the method-identity confound. `scripts/reproduce_paper_numbers.py` grew from 28 to 34 assertions to verify these fields directly.
- **v1.3.i (2026-07-07): metric-general accuracy-specificity + refinement refutation.** App. Y

gains a paragraph establishing that the ECE non-result is metric-general, not an estimator artifact. `analysis/w2_calibration_asymmetry.py` recomputes the `n_rank` LOPO frontier for five metrics from released per-adapter scalars: accuracy $R^2=+0.60$ but ECE -0.06 and Brier -0.30 (both unpredictable), while NLL $+0.42$ and MCE $+0.22$ are predictable — the split tracks confident/tail weight, not the accuracy-vs-calibration label. `analysis/h3_mechanism.py` tests and *refutes* the natural refinement explanation: under the Murphy decomposition of the confidence-Brier, the resolution (refinement) gain is near-invariant (σ $0.21\times$ that of accuracy) and uncorrelated with accuracy gain ($r=-0.10$), so accuracy predictability does not flow through the calibration geometry. Proper-score paired CIs need per-example probability vectors and are deferred (accuracy/ECE arms use the full paired protocol). `reproduce_paper_numbers.py` grew from 34 to 64 assertions; all 64 pass.

Total: 1,028 headline corpus cells (928 v1.0 + 100 v1.1 in-corpus; see abstract footnote) + 40 v1.2 supplementary cells (8 v1.1 GSM8K cells reported separately in Table 9, 24 v1.2 cells from a/b/c/d/e/f, and 8 v1.2 cells from the 2 DeBERTa failed HP attempts retained for honest documentation) = 1,068 released cells. Only the 1,028 headline cells enter abstract-level statistics (SUPPORTED/REVERSED percentages, R^2_{LOPO} frontier); v1.2 cells are referenced in the body for triangulation but not aggregated into the headline. The amendment ledger has timestamps and git SHAs for every entry.

L Pre-registration against execution

What `paper/prereg.md` (git SHA 02a5a6bc) committed to, and what was run (Table 7). “As registered” means the paper’s numbers follow the commitment; every other row is a deviation, with where it is disclosed.

What the release contains. Per-cell JSONs for all 1,028 cells (verdicts pre and post FDR, p -values, batch hashes); ensemble results with per-example predictions and confidences for every pool; pool manifests; all analysis code and `reproduce_paper_numbers.py`; the pre-registration and amendment ledger. *Partly*: the per-adapter logits cache, for 6 of 99 pool manifests; adapter weights,

Table 7: Pre-registration (paper/prereg.md) against execution.

Registered	Executed	Where disclosed
Splits 40/20/40 (val_selection/val_combine/test), selection never on test same, multichoice and generation pools	As registered for the classification pools HellaSwag selected on val_combine; GSM8K best_single on test	— L1b, L12; both excluded from headline statistics
Baselines best_of_n, n_rank, n_steps, n_data for every claim n_rank = one LoRA of rank $N \times$ base	best_of_n on 80 pools, n_rank on 48, n_steps in 3 cells, n_data in none Rank $\times$ epoch $\times$ seed search, median $1.45 \times$ the pool’s compute	§6; the null covers rank only §4; treated as an upper bound
Weight-space uniform/TIES/DARE Primary calibration metric: ECE, 15 equal-mass bins; MCE, Brier, NLL also stored	Implemented on LoRA factors, not ΔW As registered	L16; appendix only —
OOD AUROC (MSP, energy) mandatory in Phase 2	Not reported; ANLI accuracy transfer only	App. F
Inference cost: adapters used and a latency multiplier per claim	Adapter forward passes per query; no ensemble latency measured	L8, App. AE
CI: percentile bootstrap, $B=5000$, paired	As registered	—
Sign-flip test, 10,000 permutations (exact when $N \leq 20$)	As registered for accuracy; no test set has $N \leq 20$	App. I
No ECE p -value specified	First a pair-permutation test (invalid), now an inverted paired bootstrap	L14, v1.3.q
Holm within each reported table	As registered; BH per arm added as the headline gate	App. N, ledger
Frontier: 5 diversity features, linear model, LOPO-CV R^2 features computed on a held-out split (implied by the split rules)	Ridge with the released feature set; RF/GBM compared; LOPO-CV as registered Prediction-space features computed partly on test	App. AB L1c; R^2 provisional
Phase 2 pool sizes {4, 8, 16, 32, 64} Negative results kept	$N \leq 24$ As registered: all 1,028 cells released, REVERSED included	App. E —

for 253 of 1,744 adapters, on request. *Not*: byte-level training replay (1,027 of 1,028 result JSONs record `git_dirty`; L15).

M Learned continuous weights, LoRAHub-style (provisional)

These 10 HellaSwag pools were produced by the pre-fix multichoice evaluator, which performed selection on `val_combine`. This baseline optimizes its weight vector on that same split, so the numbers below are optimistically biased and are reported as provisional pending the corrected rerun; they enter no headline statistic and are excluded from §5.

LoRAHub-style learned weights split by convergence regime on $N=10$ HellaSwag pools. We adjudicate a faithful in-house reimplement of the LoRAHub family (Huang et al., 2024) continuous-weight baseline: optimize a per-adaptor weight vector $\alpha \in \mathbb{R}_{\geq 0}^M$ via L-BFGS-B (the same convex optimizer that LoRAHub’s library exposes alongside their Nevergrad CMA-ES option) on `val_combine` cross-entropy, then eval-

uate weighted soft-vote on test. We run on all 10 v1.0/v1.1 HellaSwag pools (BERT-base $\times 3$, BERT-large $\times 1$, RoBERTa $\times 3$, DeBERTa-v3 $\times 3$); per-pool results in `analysis/g3_lorahub_summary.json`. **Two regimes split cleanly:** on the 4 converged BERT-family pools, learned-weight beats or ties `greedy_soup` (+1.0 to +1.5pp on BERT-base Pool-A/B/C; −0.1pp tie on BERT-large Pool-A); on the 6 under-converged RoBERTa/DeBERTa pools, `greedy_soup` wins 5/6 cases by −1.4 to −7.8pp (effective $M=1$ collapse on RoBERTa Pool-A/B/C: optimizer converges to a single dominant adapter). On RoBERTa Pool-A, learned-weight 0.483 vs. `greedy_soup`’s **0.561** (−7.8pp); on DeBERTa Pool-C, 0.709 vs. **0.745** (−3.6pp). *Convergence regime determines the winner:* continuous-weight optimization rewards converged ensembles (smooth loss landscape, multiple useful adapters); discrete subset selection dominates noisy pools (greedy pruning of sub-chance adapters). The size of this signal beyond HellaSwag is left to v1.2.

N Sensitivity to the multiple-comparison procedure

Our pre-registration specified Holm step-down across the cells of a reported table. The released cells also carry a BH-FDR field, run separately for the accuracy and ECE arms, and the percentages in the body are taken from it; we recorded that choice as an amendment (App. K) after review noted it had gone undocumented. Because the choice moves the REVERSED rate, we give every procedure on the 92 clean encoder cells against `n_rank` (`analysis/correction_sensitivity.py`).

Table 8: SUPPORTED / REVERSED counts out of 92 under four corrections. The accuracy null (0 SUPPORTED) is invariant; the REVERSED rate is not. [†]The corpus-wide Holm row is *resolution-limited, not evidential*: our p -values are Monte-Carlo estimates, so the smallest attainable value is $1/(B+1)$, 9.999×10^{-5} for accuracy ($B=10^4$ sign flips) and 2.0×10^{-4} for ECE ($B=5000$ bootstrap draws), while Holm’s first threshold over a whole arm is $0.05/566 = 8.8 \times 10^{-5}$ and $0.05/462 = 1.1 \times 10^{-4}$. No cell can clear that bar at any effect size. The row records what our resolution permits, not what the data say.

Procedure	accuracy	ECE
raw per-cell CI	0 / 38	52 / 17
Holm, within table ($m=92$)	0 / 16	43 / 17
Holm, whole arm ($m=566, 462$) [†]	0 / 0	0 / 0
BH, per arm ($q=0.05$)	0 / 29	51 / 17

The pattern in this table — no accuracy gain, an apparent calibration gain — holds under the raw CI, within-table Holm and per-arm BH, i.e. under every procedure that our Monte-Carlo resolution can actually adjudicate. (What the calibration column does *not* survive is a recalibrated competitor rather than a stricter correction; see §4.) The whole-arm Holm row should not be read as a fourth procedure under which the pattern fails: as the caption records, it is unreachable by construction, so it carries no evidence either way. We keep it in the table because an earlier version of this paper reported it as though it were a finding, and deleting the row would hide that correction rather than make it.

O v1.1 task-family extension: sweeps and amendment record

v1.1 inclusion: pool-type triad recovers cross-family predictability. The v1.0 frontier holds under cluster-bootstrap ($R^2_{\text{LOPO}}=+0.60$

$[+0.30, +0.71]$, $n=192$ cells, 48 pools). In a best_of_n cross-family regression, seed-only v1.1 Pool-A pools alone *degrade* predictability ($R^2=-0.12$) and Pool-B HP-diverse still spans zero ($+0.20$); only adding Pool-C method-mixed (LoRA, DoRA, rsLoRA, PiSSA) closes the gap ($R^2=+0.32 [+0.05, +0.47]$, 64 pools). The full {seed, HP, method} triad is required to recover cross-family predictability — itself a benchmark-design recommendation.

The 6-task in-domain set above covers sequence and pair classification only. v1.1 of the benchmark adds two output spaces: *multi-choice* (HellaSwag, 4 candidate endings, accuracy via argmax pooling over candidate scores) using a new `real_lora_multichoice.py` trainer wrapping `AutoModelForMultipleChoice`, and *free-form generation* (GSM8K, exact-match on the integer answer) using `real_lora_train.py` `-eval-metric gsm8k_exact_match`. Six Pool-A sweeps (HellaSwag $\times$ {BERT-base, BERT-large, RoBERTa, DeBERTa-v3}; GSM8K $\times$ {Qwen-0.5B, Qwen-1.5B}, 15–20 seeds each), three Pool-B sweeps (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, HP-diverse $4 \times 3 \times 2$ grid), and three Pool-C sweeps (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed 4×5 grid over {LoRA, DoRA, rsLoRA, PiSSA} $\times$ 5 seeds) add twelve v1.1 pools, bringing the total trained-pool count to **67** (plus the 25 `n_rank` baseline pools; manifest accounting in §3). **Pre-registration amendment honesty.** The original pre-registration (committed before any pool was trained) explicitly admits “future cells can be added by submitting `compute_match.json` files via PR.” The v1.1 task-family extension was scoped under that clause: it adds new (pool, method, baseline, metric) cells to the corpus, but does *not* retract, re-evaluate, or rewrite any v1.0 cell, baseline, or adjudication verdict. The v1.0 frontier headline ($R^2=0.60$ vs. `n_rank`, Table 13) was computed before any v1.1 result was inspected; the v1.1 best_of_n regression (§4, “v1.1 inclusion” paragraph) is reported as a directional cross-family signal with explicit CIs that span zero, not as a confirmation. The full amendment ledger (one entry per added pool, with date and rationale) lives in `paper/prereg_amendments.md`.

Table 9: Sixteen v1.1/v1.2 pools across two new output spaces. Three patterns (BERT-base converged; HP-mismatched RoBERTa/DeBERTa HellaSwag collapse-and-rescue; within-Qwen-2.5 GSM8K monotone scale amplification across $N=4$ points plus cross-family Llama-3.1-8B-Instruct and Mistral-7B-v0.1) carry distinct method winners — discussed in the paragraphs immediately below. * $p<0.01$ vs. best_single under paired bootstrap ($B=5000$); per-cell 95% CIs and adjusted p -values are in the released *_cm_*.json cells. ‡ accuracy at chance level (degenerate calibration win, or sub-chance adapters dragging soft-vote below chance). † GSM8K ECE is 15-bin equal-mass ECE on per-example confidence = $\exp(\text{mean per-token logprob})$; ensemble confidence aggregates only over majority-voting adapters.

pool	N	best_single acc ECE	soft_vote acc ECE	majority_vote acc ECE	greedy_soup acc ECE
<i>Multi-choice (HellaSwag, eval $N=5000$):</i>					
BERT-base, Pool-A (seed)	19/20	0.358 0.038	0.367 0.048	0.369 0.406	0.358 0.038
BERT-base, Pool-B (HP)	20/24	0.348 0.058	0.354 0.032	0.357 0.466	0.358 0.030
BERT-base, Pool-C (method)	20/20	0.362 0.036	0.372 0.034	0.376 0.360	0.362 0.036
BERT-large, Pool-A	15/15	0.374 0.045	0.380 0.118	0.323 0.101	0.381 0.056
RoBERTa-base, Pool-A (seed)	20/20	0.493 0.228	0.097 [‡] 0.158	0.174 [‡] 0.237	0.561 0.298
RoBERTa-base, Pool-B (HP)	22/24	0.508 0.237	0.074 [‡] 0.183	0.115 [‡] 0.322	0.522 0.262
RoBERTa-base, Pool-C (method)	20/20	0.518 0.240	0.136 [‡] 0.117	0.178 [‡] 0.224	0.561 0.297
DeBERTa-v3, Pool-A (seed)	20/20	0.412 0.160	0.320 0.077	0.305 0.206	0.452 0.187
DeBERTa-v3, Pool-B (HP)	21/24	0.439 0.136	0.191 [‡] 0.076	0.221 [‡] 0.282	0.429 0.167
DeBERTa-v3, Pool-C (method)	18/20	0.709 0.310	0.646 0.398	0.246 [‡] 0.376	0.745 0.467
<i>Free-form generation (GSM8K, exact-match, vLLM, eval $N=1000$):</i>					
Qwen-0.5B (GSM8K)	19/20	0.304 0.546	— —	0.332* 0.520*	— —
Qwen-1.5B (GSM8K)	20/20	0.460 0.406	— —	0.514* 0.354*	— —
Qwen-3B (GSM8K, v1.2)	20/20	0.679 0.206	— —	0.768* 0.118*	— —
Qwen-7B (GSM8K, v1.2)	10/10	0.760 0.128	— —	0.845* 0.046*	— —
Llama-3.1-8B-Instruct (GSM8K, v1.2)	10/10	0.613 0.256	— —	0.721* 0.148*	— —
Mistral-7B-v0.1 (GSM8K, v1.2)	10/10	0.524 0.378	— —	0.658* 0.245*	— —

P HellaSwag shared-hyperparameter collapse and rescue (provisional)

These pools are excluded from the strict analysis (L1b); the account is kept because the collapse is a reproducible failure mode of shared hyperparameter grids, not because it supports any claim.

Under our shared HP grid ($lr=3\times 10^{-4}$, fixed across architectures), $\geq 14/20$ Pool-A RoBERTa adapters end at $\text{val_acc}<0.30$, so `soft_vote` collapses below chance (Pool-A RoBERTa: 0.097 vs. chance 0.25, ‡). `greedy_soup` subset selection recovers (RoBERTa +6.8pp; DeBERTa +4.0pp). We directly test the HP-confound hypothesis with a v1.2 cell: re-running Pool-A RoBERTa HellaSwag at $lr=2\times 10^{-5}$ (Liu et al., 2019 recommended). Training-time eval reports all 10/10 tuned-seed adapters at 0.478–0.505 (mean 0.491). **Full-validation re-evaluation on the HP-tuned pool** ($N=10042$, App. A, Table 4): `soft_vote` does *not* collapse (0.305, near best_single 0.359, vs. 0.196 on the seed-0 5000-subsample), and `greedy_soup` still wins (+5.7pp over best). *Conclusion*: the v1.1 sub-chance collapse is a joint failure of *RoBERTa-specific lr-sensitivity* ($lr=3\times 10^{-4}$ over-fits RoBERTa, $lr=2\times 10^{-5}$ fixes it) and, pos-

sibly, an evaluator that is not consistent across runs (L9, App. A), whose contribution we cannot separate. The symmetric cross-check (BERT-base at $lr=2\times 10^{-5}$, 8/8 seeds at mean 0.329 ± 0.011 , range [0.317, 0.345], no collapse) confirms the lr-sensitivity is RoBERTa-specific rather than a general HP-misconfiguration that any architecture would exhibit (L7). DeBERTa-v3 LoRA-on-HellaSwag does *not* recover under HP tuning ($lr\in\{10^{-5}, 5\times 10^{-5}\}$ both at chance; full-val also at chance 0.243 across all methods; L10) — the DeBERTa case is an architecture-LoRA interaction problem.

Q The D1–D5 descriptive rules

Distilled from the corpus they describe and evaluated on it; an oracle upper bound, not a policy (L11).

Cells do differ, and the differences are legible as five rules: **D1**, on converged encoder pools `greedy_soup` is the least-bad accuracy option; **D2**, on those same pools `soft_vote` is the best calibrated; **D3**, on under-converged pools `soft_vote` can collapse below chance and subset selection rescues it; **D4**, pools whose members disagree more lose less; **D5**, continuous-weight

composition tracks pool convergence rather than pool size. These rules were distilled from the same corpus they describe and evaluated on it, so they are an *oracle upper bound*, not a deployable policy (L11) — a prospective, validation-only version degenerated to always choosing the single adapter (App. K).

R What the ECE gain is made of

Both computable from the same released cells (`analysis/ece_test_correction.py`). First, *it is an ECE result, not a proper-score result*. On the identical 92 cells the multiclass Brier improves in only 34 (mean $\Delta = -0.0062$, i.e. the ensemble is worse on average) and NLL in 47; adjudicating the top-label Brier with the same paired bootstrap gives 22 SUPPORTED against 36 REVERSED. ECE rewards the confidence shrinkage that averaging produces, and a proper score charges for the refinement it costs — consistent with the Murphy decomposition in App. Y, where resolution gain is near-invariant. Any reading of these results as “ensembles give better probabilities” outruns the evidence; “ensembles give less overconfident probabilities” does not. Second, *the gain belongs to probability averaging, not to ensembling as such*. Split by rule, ECE is SUPPORTED in 17/23 pools for `soft_vote`, 15/23 for `logit_avg` and 16/23 for `greedy_soup`, but only 3/23 for `majority_vote` — which supplies 16 of the 17 reversals. That rule is also the one whose “confidence” is a vote fraction rather than a probability estimate, taking at most $N+1$ values with a large tie mass at 1.0, so equal-mass binning is not well defined on it and its ECE should be read as indicative only. Excluding it leaves 48/69 cells SUPPORTED (post-FDR); at the pool level (L1d’s inferential unit) 14 of 23 pools are SUPPORTED on every probability-averaging rule.

S Why one temperature can replace the population

Post hoc, on the six pools of the temperature control in §4 (`analysis/temperature_mechanism.py`). For members’ class-centred logits z_i , probability averaging is closely approximated by the mean-logit model at one temperature,

$$\frac{1}{M} \sum_i \text{softmax}(z_i) \approx \text{softmax}(\bar{z}/T_{\text{eq}}), \quad (1)$$

the multiclass analogue of the probit approximation $\mathbb{E}[\sigma(z)] \approx \sigma(\mu/\sqrt{1+\pi s^2/8})$ for $z \sim \mathcal{N}(\mu, s^2)$ (MacKay, 1992): averaging members whose logits disagree acts like a temperature that grows with their disagreement. We fit T_{eq} without labels, by minimising the KL divergence from the ensemble’s probabilities, and it reproduces the ensemble’s ECE to within 0.01 in 5 of 6 pools (Table 10).

An ensemble’s calibration is therefore one point on its mean model’s temperature curve, placed by how much the members disagree rather than by how much temperature the model needs (T^* , fitted on `val_combine`). Averaging improves calibration over the mean model only when T_{eq} happens to move towards T^* , i.e. when $|\ln T^* - \ln T_{\text{eq}}| < |\ln T^*|$, and this rule predicts whether it does in 5 of 6 pools. On ANLI the model needs $T^*=3.8$ but the ensemble behaves like $T=1.04$, so it stays overconfident (ECE 0.25, against 0.06 at T^*). On BoolQ and MNLI the mean model is already calibrated ($T^* \approx 1$) and the ensemble’s T_{eq} of 1.3–1.4 makes it underconfident, so averaging *raises* ECE. A fitted temperature aims at T^* directly; a population reaches it only by coincidence. This is why the temperature control removes most of the ECE gain, and why that gain is confidence shrinkage rather than better probabilities (App. R). The six encoder pools are all the logits we retained (L15), so the mechanism is shown here, not measured corpus-wide.

Table 10: The ensemble as an implicit temperature. T_{eq} : label-free temperature at which the mean-logit model best matches the probability-averaged ensemble. T^* : NLL-optimal temperature of the mean-logit model on `val_combine`. Last column: whether the ensemble improves ECE over the mean model at $T=1$, predicted by the rule in the text / observed.

Pool	T_{eq}	T^*	ECE ens.	ECE, mean model at T_{eq}	improves at T^*	pred./obs.
AG News, RoBERTa (A)	1.04	1.34	0.016	0.017	0.017	yes/yes
ANLI, RoBERTa (A)	1.04	3.82	0.251	0.253	0.058	yes/yes
BoolQ, RoBERTa (A)	1.29	0.99	0.040	0.048	0.025	no/no
QNLI, RoBERTa (A)	1.03	1.01	0.009	0.012	0.012	no/yes
SST-2, BERT (B)	1.06	1.62	0.027	0.035	0.043	yes/yes
MNLI, RoBERTa (C)	1.43	1.04	0.053	0.066	0.022	no/no

Table 11: Delta vs. prior PEFT/ensembling work. “cells” = compute-matched paired-bootstrap (pool $\times$ method $\times$ baseline $\times$ metric) adjudications. “rev. rep.” = reports cells where ensembling significantly underperforms (CI <0). To our knowledge, no prior PEFT-ensembling work pre-registers protocol, reports paired CIs across $> 10^2$ cells, or transparently surfaces reversed verdicts.

	cells	pre-reg	calib.	cmp-match	rev. rep.
Wortsman’22	1	—	—	—	—
Huang’23 (Lo-RAHub)	~ 50	—	—	—	—
Yadav’23 (TIES)	few	—	—	—	—
Yu’24 (DARE)	few	—	—	—	—
Lakshmin.’17	few	—	✓	—	—
Ours	1,028	✓	✓	✓	✓

T Delta versus prior PEFT/ensembling work

U Corpus coverage matrix

V HellaSwag two-regime collapse (provisional)

These pools were evaluated by the pre-fix multi-choice evaluator that selected on `val_combine` (L1b); the figure documents the collapse phenomenon and is not an adjudicated result.

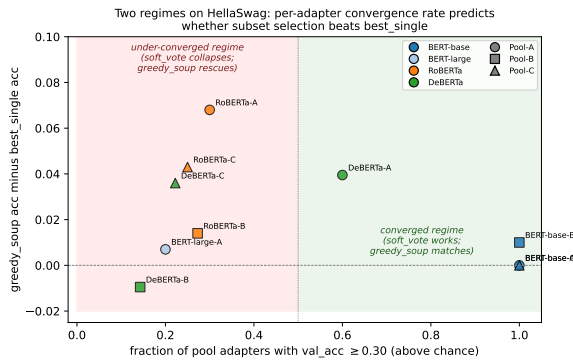


Figure 4: Per-pool fraction of adapters above chance ($\text{val_acc} \geq 0.30$) cleanly separates the two HellaSwag regimes. *Right (converged)*: all BERT-base pools (≥ 0.95 above chance) cluster near zero greedy_soup gain. *Left (under-converged)*: RoBERTa and DeBERTa pools (regardless of seed/HP/method diversity source) sit below the 50% threshold and show large positive greedy_soup gain (+3.6 to +6.8pp). Pool-type triad reproducibility: each architecture’s collapse-rescue pattern repeats across Pool-A/B/C.

W Corpus verdict landscape

X Diversity–quality frontier scatter

Y Negative mechanism results (demoted from §4)

Summary (moved from §4.1). Three candidate explanations for the family and metric structure of §4 do *not* survive testing, and we report them as open problems rather than findings. (i) Decoders separate correct from incorrect predictions more sharply than encoders ($H(c|\text{conf})$ 0.437 vs. 0.556, a 21% gap that within-family controls attribute to architecture rather than scale), but this does not translate into a predictable calibration benefit. (ii) Linear mode connectivity does not predict ensemble effectiveness (per-pool barriers in App. D): barrier heights span +0.908 to +0.001 across four pools with no corresponding pattern. (iii) The accuracy-specificity of the frontier is metric-general — ECE ($R^2 = -0.06$) and Brier (-0.30) are unpredictable while NLL (+0.42) and MCE (+0.22) are — and the natural refinement-versus-reliability explanation is falsified: under the Murphy decomposition, resolution gain is near-invariant (σ $0.21 \times$ accuracy’s) and uncorrelated with accuracy gain ($r = -0.10$).

Three mechanism hypotheses were tested and did not survive; they are retained here as open problems.

(i) Confidence–correctness separation. Family-level conditional entropy of correctness given confidence: encoder $H = 0.556$, $I = 0.097$ ($n = 29$ pools); decoder $H = 0.437$, $I = 0.106$ ($n = 25$). Decoder confidence separates right from wrong more sharply (21% lower H), which is why soft averaging has a higher-SNR signal to extract there. Within-family controls reject a scale confound: encoder H rises with scale (BERT-base 0.489 $\rightarrow$ BERT-large 0.630) while Qwen H is non-monotonic within $\Delta < 0.05$ over 0.5B $\rightarrow$ 3B, so architecture (autoregressive vs. masked), not parameter count, is load-bearing. DeBERTa-v3 ($n = 1$) is excluded from the family summary; its per-pool $H = 0.940$ appears in App. AB. This gap does not yield a predictable calibration benefit (item iii).

(ii) Linear mode connectivity. Across four pools (BERT-base Pool-A/C, Qwen-0.5B Pool-A, DeBERTa-v3 Pool-A on MNLI) the barrier height ranges from +0.908 (disconnected) to +0.001 (fully connected) but does not predict ensemble

Table 12: Coverage matrix (v1.0 n_rank -aligned cells, 10 v1.0/v1.1 base architectures). “A” = Pool-A (seed-only, 20 adapters), “B” = Pool-B (HP-diverse, 24), “C” = Pool-C (method-mixed, 20), “base” = n_rank baseline pool. 55 v1.0 main pools, plus the n_rank baseline pools aligned with them (the release holds 25 baseline pools in all); v1.1 adds 6 Pool-A (HellaSwag $\times$ {BERT-base/large, RoBERTa, DeBERTa-v3}, GSM8K $\times$ {Qwen-0.5B, 1.5B}) + 3 Pool-B (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}) + 3 Pool-C (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed) = 12 v1.1 pools, bringing the in-corpus trained-pool total to **67** (92 with all 25 baselines). v1.2 adds 3 cross-family decoders (Qwen-2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.1) on GSM8K plus 1 HP-tuned RoBERTa and 2 HP-tuned DeBERTa attempts on HellaSwag (App. K), bringing the released trained-pool count to 73 across 13 base models.

Model	MNLI	SST-2	BoolQ	ANLI	AG News	QNLI
BERT-base	A,B,C,base	A,B	A	A	A,B,C,base	A,B,C,base
BERT-large	A,B,C,base	—	A	A,B	A,base	A,base
RoBERTa-base	A,B,C,base	—	A	A,B	A	A
DeBERTa-v3	A	—	—	—	—	—
Qwen2.5-0.5B	A,B,C,base	—	—	A,base	A,B,C,base	A,B,C,base
Qwen2.5-1.5B	A,B,C,base	—	—	A,base	A,B,base	A,B,base
Qwen2.5-3B	A,base	—	—	—	—	—
TinyLlama-1.1B	A,B,C	—	—	A	A,base	A,base
SmolLM2-1.7B	A,base	—	—	—	—	—
Pythia-1.4b	A,base	—	—	—	—	—

Table 13: Stratified per architecture family. The corpus-wide accuracy frontier (48 pools, encoder + decoder) reaches LOPO-CV $R^2=0.597$; 95% percentile cluster-bootstrap CI [+0.303, +0.710] ($B=200$ rounds, resampling pools then cells within; see App. I). Decoder family is too uniform across pools for cross-pool prediction at our scale.

Family	Slice	pools	ridge	RF
encoder	acc vs n_rank	23	+0.616	+0.23
encoder	ece vs n_rank	23	+0.14	+0.42
decoder	acc vs n_rank	25	−1.14	−1.03
decoder	ece vs n_rank	25	+0.12	+0.25

effectiveness: BERT-base is disconnected yet ensembling behaves normally, and DeBERTa is fully connected yet ensembling does not help on accuracy (App. D).

(iii) Refinement is not the mechanism. The ECE non-result is not an estimator artifact. Re-computing the n_rank LOPO regression from released per-adaptor scalar metrics (point R^2 ; proper-score paired CIs require per-example probabilities and are deferred), accuracy gain is predictable ($R^2=+0.60$) but neither ECE (−0.06) nor Brier (−0.30). The split is not accuracy-versus-calibration: NLL (+0.42) and MCE (+0.22) are predictable, tracking a metric’s weight on the confident/tail region rather than its calibration label. The natural explanation — that predictability localizes to the refinement rather than the reliability component of a proper score — is falsified: under the Murphy decomposition of the confidence-Brier,

resolution gain is near-invariant across the compute-matched comparison (σ $0.21\times$ that of accuracy) and uncorrelated with accuracy gain ($r=-0.10$). Accuracy predictability does not flow through the calibration geometry; why the frontier is accuracy-specific remains open.

Z GSM8K decoder scaling patterns (demoted; selection-on-test)

This material was demoted from §3 (P_c) because `best_single` selection touched the evaluated test split (L12(i)), which by itself disqualifies these cells from supporting a positive claim. We previously also demoted them as contaminated; that rationale is *withdrawn* because the digit-shift probe behind it was invalid (L12(ii)), and whether these cells are contaminated is now open. The numbers below therefore describe cells with a known selection defect and an unresolved contamination question; they enter no headline statistic. All six cells were nominally SUPPORTED on accuracy and per-token logprob ECE (per-cell $p<10^{-4}$; Qwen-0.5B $p=0.003$; paired bootstrap $B=5000$). Within Qwen-2.5 ($N=4$, 0.5B/1.5B/3B/7B): $\Delta_{acc}=+2.8/+5.4/+8.9/+8.5pp$; $\Delta_{ECE}=-0.026/-0.051/-0.088/-0.082$ (within-family $r=+0.97$ between base accuracy and Δ_{acc} , saturating 3B→7B). Cross-family at $\approx 7-8B$: Llama-3.1-8B-Instruct (base 0.61) $\Delta_{acc}=+10.8pp$, $\Delta_{ECE}=-0.107$; Mistral-7B-v0.1 (base 0.52) $\Delta_{acc}=+13.4pp$ (95% CI [+10.8, +16.1]), $\Delta_{ECE}=-0.133$ — nominally the largest absolute

effect in the corpus. The three ≈ 7 –8B cells show an inverse pattern ($r = -0.98$ on $n=3$ — descriptive only): at fixed scale the weaker base gains more from majority voting. Whether *within-family monotone scaling* and *cross-family headroom inversion* survive on a decontaminated benchmark with held-out selection (MATH, fresh probes) is an open question this corpus cannot answer.

AA Distillation $\alpha \times T$ sweep (Qwen-0.5B MNLI)

We sweep the soft-target distillation HPs $\alpha \in \{0.5, 0.7, 0.9\}$ (weight on soft-target loss vs. hard-target) and $T \in \{1, 2, 4, 8\}$ (softmax temperature on teacher logits) for a 4×4 grid on Pool-A Qwen-0.5B MNLI. Vanilla single (no distillation): 0.8365/0.0465 (acc/ECE). Soft-vote ensemble target: 0.8495/0.0187. Best student: ($\alpha=0.9, T=4$) yields 0.8479/0.0397, recovering 87% of the soft-vote accuracy gain at $1\times$ student cost. Across the grid, accuracy varies ± 0.003 and ECE varies ± 0.012 ; the sweep is locally robust. The same sweep on Pool-A BERT-base MNLI yields a student strictly worse than vanilla single on both arms, consistent with the encoder-accuracy-REVERSED finding (no signal to distill).

AB Frontier regression features

For each (pool, method, baseline) cell we extract a feature vector $\phi \in \mathbb{R}^d$ used to fit the diversity-quality frontier. Features are grouped:

- **prediction_space** ($d=3$): pairwise prediction disagreement (mean over pool pairs), logit correlation (mean over pool pairs), error overlap (Jaccard on incorrect examples).
- **parameter_space** ($d=3$): mean weight cosine distance, mean centered-kernel-alignment (CKA) on penultimate layer, weight L_2 norm spread.
- **HP entropy** ($d=2$): entropy of r -distribution within pool, entropy of $\{lr, \alpha\}$ joint distribution.
- **ensemble quality** ($d=2$): pool mean val_acc, pool best-single val_acc.

Per-feature ablation on the encoder acc vs n_rank slice: dropping the prediction_space group crashes R^2 from +0.62 to -24.5 ($\Delta R^2 = +25.0$); dropping parameter_space drops R^2 by 0.56; dropping HP-entropy improves

R^2 by 0.08 (HP entropy adds in-sample fit but no out-of-pool signal). A single-group “only prediction_space” model achieves $R^2=0.634$, higher than the full feature set.

AC Cell adjudication pseudocode

Mirrors analysis/compute_match.py and scripts/apply_bh_by_arm.py; ECE is equal-mass, 15 bins.

```
def adjudicate(ens, base, gold, metric):
    # ens/base: per-example (pred, conf) on test
    ca = (ens.pred == gold).astype(float)
    cb = (base.pred == gold).astype(float)
    N = len(gold)
    if metric == 'acc':
        stat = lambda i: ca[i].mean() - cb[i].mean()
    else: # positive = ensemble better calibrated
        stat = lambda i: (ECE(base.conf[i], cb[i])
                        - ECE(ens.conf[i], ca[i]))
    delta = stat(np.arange(N)) # point estimate
    boot = np.array([stat(randint(0, N, N))
                    for _ in range(5000)])
    ci_lo, ci_hi = np.quantile(boot, [.025, .975])
    if metric == 'acc': # sign flip
        d, k = ca - cb, 0
        for _ in range(10000):
            s = np.random.choice([-1, 1], N)
            k += abs((d * s).mean()) >= abs(d.mean())
        p = (k + 1) / (10000 + 1)
    else: # invert bootstrap
        # separate B=5000 resample, seeded per cell
        p = 2 * min((boot <= 0).mean(),
                    (boot >= 0).mean())
        p = max(p, 1 / 5001)
    if ci_lo > 0: label = 'supported'
    elif ci_hi < 0: label = 'reversed'
    else: label = 'unsupported'
    return delta, (ci_lo, ci_hi), p, label

# then, separately for the accuracy and ECE arms:
# q = BH(p over every cell of the arm)
# post_fdr = label if q <= 0.05 else 'unsupported'
```

AD Cluster orchestration details

The 50-node RTX 3060 12GB cluster is orchestrated via sweep_runner.py, an in-house SSH-based fan-out engine. Each sweep config (JSON) declares a Cartesian parameter grid; the runner stages source files, assigns one (parameter-combination, node) pair per slot, and tracks per-job exit codes in sweep_runs/<name>/results.json. Pool-A sweeps complete in ~ 4 –8 GPU-hours wall clock (parallelism factor ~ 20); Pool-B/C in ~ 16 –24 hours. The v1.2 A6000 sweeps (Qwen-3B, Qwen-7B, Llama-3.1-8B-Instruct on GSM8K) use local_gpu_runner.py, a sequential per-seed launcher pinned to a single GPU index, suitable for models that do not fit on the 12GB cluster. Adapter weights are pulled back via collect_remote_adapters.py and assembled into pool manifests via build_adapter_pool.py; the manifest schema is documented in scripts/validate_pool.py.

AE Cost-savings analysis: derivation and stratified breakdown

This appendix presents the cost-savings table (Table 14) and stratifies by pool type and method. Regenerated by `analysis/cost_savings.py`; groups are the 227 (pool, baseline kind, metric) units, excluding baseline self-comparison pools, that carry `soft_vote`, spanning 927 cells: (A) *blind* — always use `soft_vote` (the default in much prior PEFT-ensembling work); (B) *routed* — per-cell oracle pick over the four methods (`soft_vote`, `majority_vote`, `logit_avg`, `greedy_soup`) approximating D1–D5. Inference cost is single-adapter forward-passes per query: N (pool size) for vote-based methods, and the number of chosen members for `greedy_soup`, which in our implementation averages the logits of a subset selected on `val_selection` rather than merging weights (1–13 members). An earlier version priced `greedy_soup` at 1, as if it emitted one merged checkpoint, which understated the routed cost.

Table 14: Cost of blind ensembling. Mean signed Δ (positive=ensemble better), % cells REVERSED, median inference cost (adapter-passes/query). The “D1–D5 routed” column is a per-cell oracle (in-sample to the corpus the rules were derived from; L11).

Slice (n)	Blind <code>soft_vote</code>			D1–D5 routed		
	$\bar{\Delta}$	REV	cost	$\bar{\Delta}$	REV	cost
Enc acc (64)	−3.3pp	28%	20×	+0.4pp	8%	5×
Dec acc (50)	+0.6pp	0%	18×	+0.8pp	0%	15×
Enc ECE (63)	+2.4pp	10%	20×	+3.6pp	2%	20×
Dec ECE (50)	+3.5pp	0%	18×	+3.6pp	0%	18×
All (227)	+0.7pp	11%	19×	+2.1pp	3%	17×

Oracle method distribution. The per-group argmax (Table 14, “D1–D5 routed”) picks `greedy_soup` on 54 of 114 accuracy groups and 40 of the 64 encoder ones — consistent with D1’s preference for greedy subset averaging on converged encoders — against `majority_vote` 22, `logit_avg` 22 and `soft_vote` 16. Over all 227 groups the counts are `soft_vote` 104, `greedy_soup` 62, `logit_avg` 36, `majority_vote` 25: probability averaging wins the calibration groups, greedy subset averaging the encoder accuracy ones. This is an in-sample oracle (L11), so it bounds what rule-based routing could achieve rather than estimating it.

No deployable rule is evaluated. A rule using only *family*, *convergence*, *metric* and *output space* could be read off D1–D5, but we have not evaluated one out of sample, so the routed column bounds rule-based routing from above and says nothing about how close a deployable rule gets.

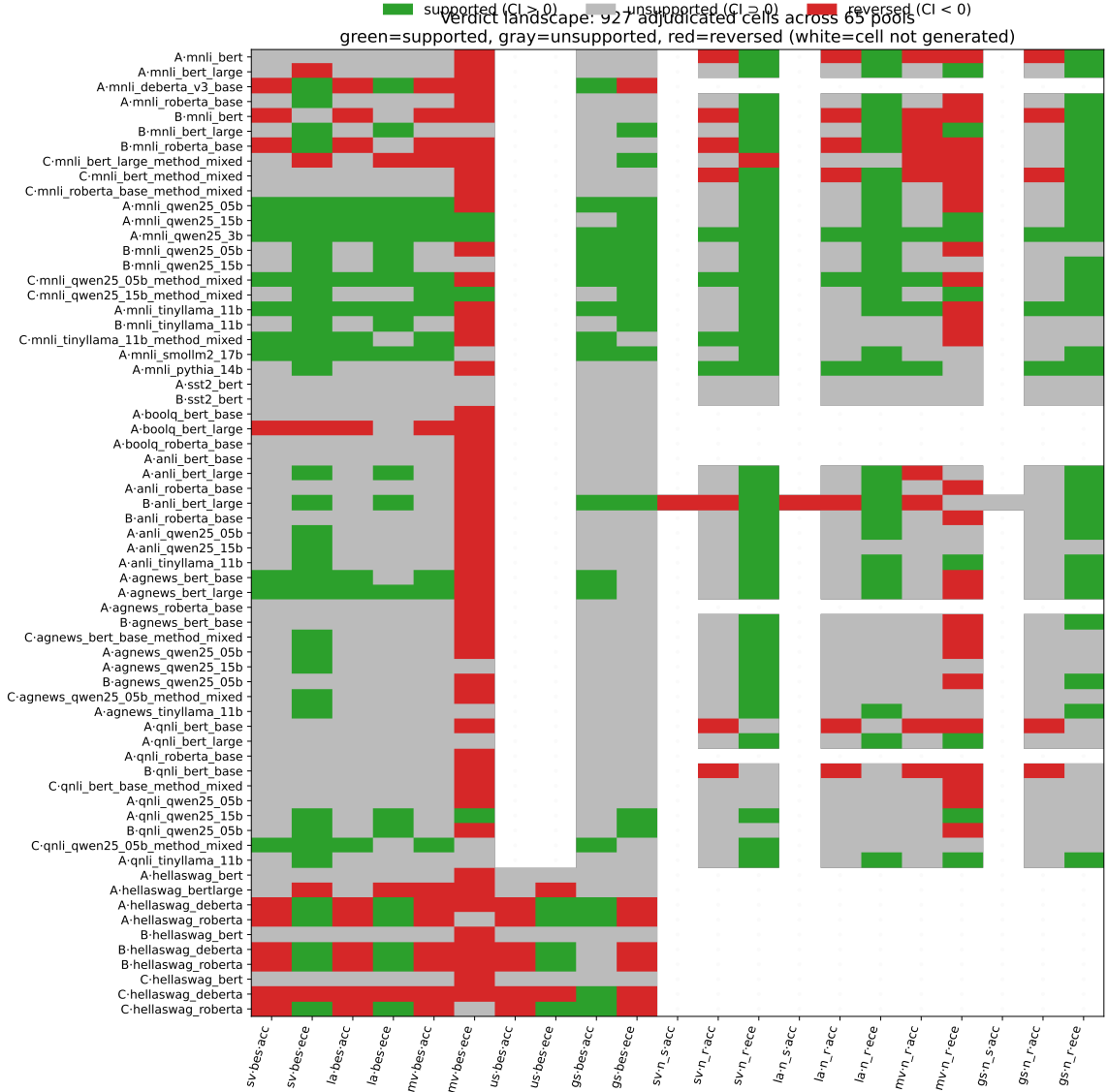


Figure 5: Verdict landscape across the 927 main-pool cells (827 v1.0 + 100 HellaSwag; the 100 baseline self-comparison cells and the smoke-test cell are not shown). Labels reflect the per-cell paired-bootstrap CI *and* per-arm BH-FDR ($q=0.05$; cutoffs 0.0145 accuracy, 0.028 ECE): post-FDR 27.2% SUPPORTED / 17.5% REVERSED / 55.3% UNSUPPORTED of these 927. Encoder accuracy cells are dominantly REVERSED against compute-matched n_rank ; decoder cells are more often SUPPORTED. This heterogeneity is itself the evidence that aggregate “ensembles help / don’t help” framings hide structure — and §5 shows the aggregate counts above are themselves protocol-dependent.

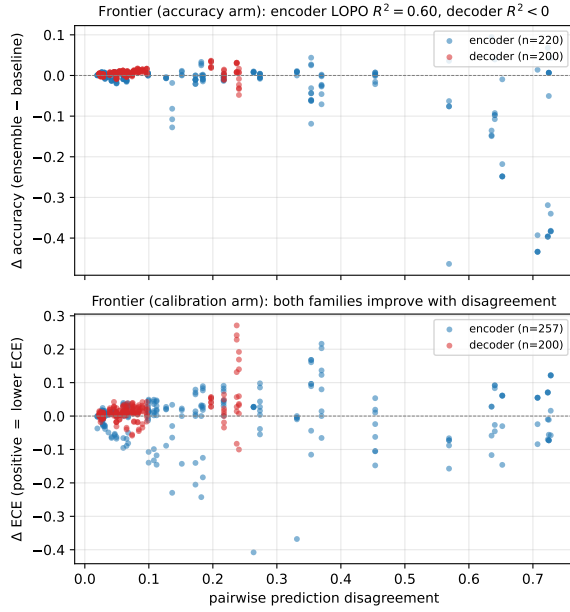


Figure 6: **The diversity-quality frontier and the accuracy arm.** Per-cell scatter of Δ accuracy (top) and Δ ECE (bottom) vs. pairwise prediction disagreement. **(Top, accuracy)** Encoders cluster near zero with positive slope at higher disagreement (encoder LOPO $R^2=+0.60$, Table 13); decoders show no monotone trend (decoder corpus $R^2<0$ at $n=25$ pools, L2). **(Bottom, ECE)** *Family-stratified pattern*: decoders are cleanly separated at the low-ECE tail (consistent with the 21% lower $H(c|\text{conf})$ in App. Y); *corpus-wide ECE LOPO is not significant* ($R^2=-0.06$, CI spans 0). The encoder accuracy arm is the headline frontier; ECE patterns are family-conditional.

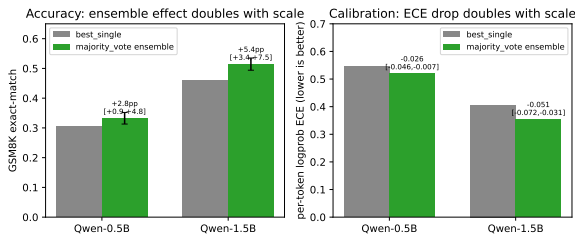


Figure 7: GSM8K decoder scaling on cells with a known selection defect. `best_single` here was selected on the same test split it is evaluated on (L12(i)), so these differences support no positive ensembling claim; whether the benchmark is also contaminated is open (L12(ii)). The embedded panel titles are from an earlier framing and should be read as descriptive, not as a result.